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APUL

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PHYSICS

1) A small object is placed 10 cm away infront of a plane mirror. If you stand behind the object 30 cm away from the mirror and look at its image, the distance focused for your eye will be-

(1) 60 cm
(2) 20 cm
(3) 40 cm
(4) 80 cm

2) Find the diameter of the image of moon formed by a spherical concave mirror of focal length 7.6 m kept on earth. The diameter of the moon is 3450 km and the distance between the earth and the moon is 3.8×10^5 km :-

(1) 6.9 cm
(2) 69 cm
(3) 3.45 cm
(4) 34.5 cm

3) A ray of light passes from vacuum into a medium of refractive index μ. If the angle of incidence is twice the angle of refraction, then the angle of incidence is :-

(1) $\cos^{-1}(\mu/2)$
(2) $\sin^{-1}(\mu/2)$
(3) $2\cos^{-1}(\mu/2)$
(4) $2\sin^{-1}(\mu/2)$

4)

An object of size 3 cm is placed 14 cm in front of a convex mirror of focal length 21 cm. Size of the image when object is moved farther from the mirror is

(1) Increasing
(2) Decreasing
(3) Same
(4) First decreasing then increasing

5) A tank is filled with benzene to a height of 120 mm. The apparent depth of a needle lying at a bottom of the tank is measured by a microscope to be 80 mm. The refractive index of benzene is :-

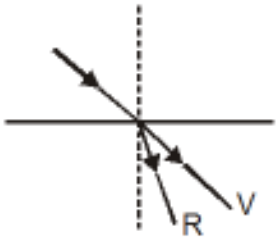
(1) 1.5
(2) 2.5
(3) 3.5

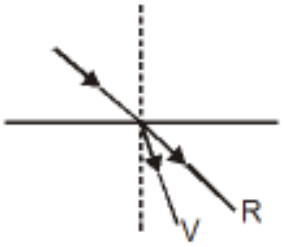
(4) 4.5

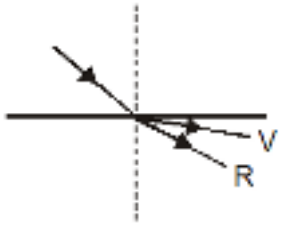
6) Total internal reflection of a ray of light is possible when the ray goes from

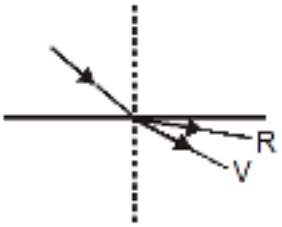
(1) denser to rarer medium and the angle of incidence is greater than the critical angle
(2) denser to rarer medium and the angle of incidence is less then the critical angle
(3) rarer to denser medium and the angle of incidence is greater than the critical angle
(4) rarer to denser medium and the angle of incidence is less then the critical angle

7) A ray of light is coming from air to water. Which of the following figure shows correct dispersion of light ?

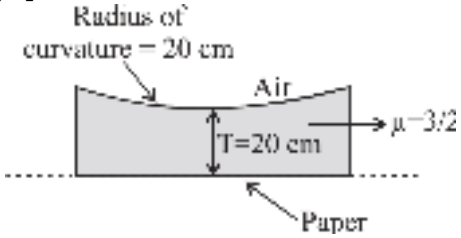
(1)


(2)


(3)


(4)


8) A plano-concave lens is placed on a paper on which a flower is drawn. how far above its actual



position does the flower appear to be ?

(1) 10 cm
(2) 15 cm
(3) 50

(4) None

9) A hollow double concave lens is made of a very thin transparent material. It can be filled with water (refractive index μ_w) or either of two liquids L_1 or L_2 with refractive indices μ_1 and μ_2 respectively ($\mu_2 > \mu_w > \mu_1$). The lens will diverge a parallel beam of light incident on it, if it is filled with
(a) L_2 and immersed in L_1
(b) L_2 and immersed in water
(c) water and immersed in L_1
(d) air and immersed in either water or L_1 or L_2
Select correct alternative :-

(1) Only (a) and (b)

(2) Only (a) and (d)

(3) Only (a), (b) and (c)

(4) Only (b), (c) and (d)

10) A thin lens focal length f and its aperture has diameter d. It forms an image of intensity I. Now the central part of the aperture up to diameter $\frac{d}{3}$ is blocked by an opaque paper. The image intensity will change to-

(1) $\frac{I}{2}$

(2) $3\frac{I}{4}$

(3) $\frac{8}{9}I$

(4) $\frac{I}{3}$

11) The objective lens of a compound microscope produces magnification of 10. In order to get an overall magnification of 100 when image is formed at 25 cm from the eye, the focal length of the eye piece lens should be :-

(1) 4 cm

(2) 10 cm

(3) $\frac{25}{9}$cm

(4) 9 cm

12) A converging beam of rays is incident on a diverging lens. Having passed through the lens the rays intersect at a point 15 cm from the lens on the opposite side. If the lens is removed the point where the rays meet will move 5 cm closer to the lens. The focal length of the lens is

(1) –30 cm

(2) 5 cm

(3) –10 cm

(4) 20 cm

13) In astronomical telescope focal length of objective and eye piece is 120 cm and 5 cm respectively. If this telescope is used to view 100 m tall tower 3 km away then find out the angular width of final image formed by eyepiece in normal adjustment.

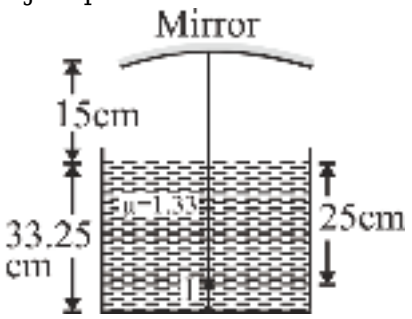
(1) $\frac{1}{30}$ radian

(2) 0.8°

(3) 30°

(4) 0.8 radian

14) A container is filled with water (μ = 1.33) upto a height of 33.25 cm. A concave mirror is placed 15 cm above the water level and the image of an object placed at the bottom is formed 25 cm below the water level. The focal length of the mirror is:-



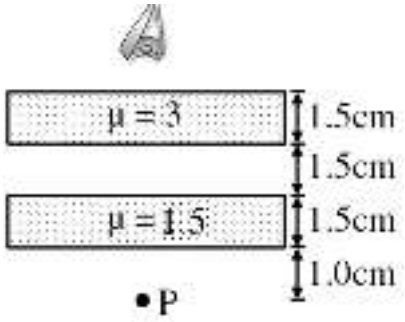
(1) 10cm

(2) 15cm

(3) 18.3cm

(4) 25cm

15) The image of point P, when viewed from top of the slabs will be :



(1) 2.0 cm above P

(2) 1.5 cm above P

(3) 2.0 cm below P

(4) 1 cm above P

16) A diverging lens with magnitude of focal length 25 cm is placed at a distance of 15 cm from a converging lens of magnitude of focal length 20 cm. A beam of parallel light falls on the diverging lens. The final image formed is :

(1) real and at a distance of 40 cm from the divergent lens

(2) real and at a distance of 6 cm from the convergent lens

(3) real and at a distance of 40 cm from convergent lens

(4) virtual and at a distance of 40 cm from convergent lens

17) In normal YDSE experiment maximum intensity is $4I_0$. In column-I, y-coordinate is given from central maxima.
In column-II resultant intensities are given. Match the two columns.

	Column-I		**Column-II**
(A)	$Y = \frac{\lambda D}{d}$	(P)	$I = I_0$
(B)	$Y = \frac{\lambda D}{2d}$	(Q)	$I = 2I_0$
(C)	$Y = \frac{\lambda D}{3d}$	(R)	$I = 4I_0$
(D)	$Y = \frac{\lambda D}{4d}$	(S)	I = zero

(1) A → R, B → S, C → P, D → Q

(2) A → S, B → P, C → Q, D → R

(3) A → P, B → Q, C → R, D → S

(4) A → Q, B → R, C → S, D → P

18) On introducing a thin sheet of mica (thickness 120 μcm) in path of the one of the interfering beams in YDSE, the central fringe is shifted through a distance equal to the spacing between successive bright fringes. If λ = 60 μcm, then find μ of the sheet. (μ = refractive index)

(1) $\frac{3}{2}$

(2) $\frac{4}{3}$

(3) $\frac{5}{3}$

(4) $\frac{8}{5}$

19) A linear aperture whose width is 0.02 cm is placed immediately in front of a lens of focal length 60 cm. The aperture is illuminated normally by a parallel beam of wavelength 5×10^{-5} cm. The distance of the first dark band of the diffraction pattern from the centre of the screen is :

(1) 0.15 cm

(2) 0.10 cm

(3) 0.25 cm

(4) 0.20 cm

20) Two polaroids are placed in the path of unpolarized beam of intensity $2I_0$ such that no light is emitted from the second polaroid. If a third polaroid is placed between these polaroids whose polarzation axis makes an angle θ with the polarization axis of first polaroid, then the intensity of

light emerging from the last polaroid will be :

(1) $\frac{I_0}{8}\sin^2 2\theta$

(2) $\frac{I_0}{4}\sin^2 2\theta$

(3) $\frac{I_0}{2}\cos^2 2\theta$

(4) $\frac{I_0}{2}\cos^4 2\theta$

21) A ray of light is incident on the surface of a glass plate at an angle of incidence equal to Brewster's angle ϕ. If μ represents the refractive index of glass with respect to air, then the angle between reflected and refracted rays is :

(1) $90 + \phi$

(2) $\sin^{-1}(\mu\cos\phi)$

(3) 90°

(4) $90° - \sin^{-1}(\sin\phi/\mu)$

22) It is found that when two waves of same intensity from two coherent sources superimpose at a certain point, then the resultant intensity is equal to the intensity of any one wave. This means that the phase difference between the two waves at that point is-

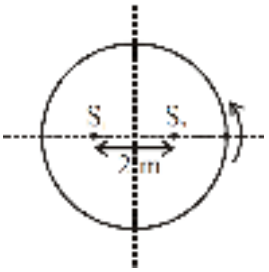
(1) 0

(2) $\frac{\pi}{2}$

(3) $\frac{2\pi}{3}$

(4) π

23) Two point sources separated by 2.0 m are radiating in phase with $\lambda = 0.50$ m. A detector moves in a circular path around the two sources in a plane containing them. How many maxima are detected?



(1) 16

(2) 20

(3) 24

(4) 32

24) A nucleus (of nuclear density ρ) disintegrates into two daughter nuclei with masses in the ratio 8 : 27. Density of the smaller nucleus is :-

(1) $\frac{2}{3}\rho$

(2) $\frac{2}{5}\rho$

(3) $\frac{8}{27}\rho$

(4) ρ

25) Consider following statements regarding nuclear force :
Statement-I : Saturation of nuclear force is the reason for the constancy of the binding energy per nucleon.
Statement-II : The nuclear binding force has to dominate over the coulomb repulsive force between protons inside the nucleus, to make the nucleus stable.
Correct choice will be :-

(1) Both Statement-I and Statement-II are correct.

(2) Statement-I is correct and Statement-II is incorrect.

(3) Statement-I is incorrect and Statement-II is correct.

(4) Both Statement-I and Statement-II are incorrect.

26) A radioactive nucleus ${}^{A}_{Z}X$ undergoes spontaneous decay in the sequence ${}^{A}_{Z}X \rightarrow {}_{Z-1}B \rightarrow {}_{Z-3}C \rightarrow {}_{Z-2}D$, where Z is the atomic number of element X. The possible decay particles in the sequence are :

(1) α, β^-, β^+

(2) α, β^+, β^-

(3) β^+, α, β^-

(4) β^-, α, β^+

27) An imaginary particle has a charge equal to that of an electron and mass 50 times the mass of the electron. It moves in a circular orbit around a nucleus of charge +e. Considering mass of the nucleus to be infinite, radius of first Bohr's orbit and energy of this shell will be :-

(1) 26.45×10^{-10} m, 0.272 eV

(2) 1.058×10^{-12} m, 0.272 eV

(3) 26.45×10^{-10} m, 0.68 keV

(4) 1.058×10^{-12} m, 0.68 keV

28) In the spectrum hydrogen atom, ratio of the longest wavelength in Lyman series to the longest wavelength in Balmer series is:

(1) $\frac{5}{27}$

(2) $\frac{1}{93}$

(3) $\frac{4}{9}$

(4) $\frac{3}{2}$

29) In a nuclear fission reaction of an isotope of mass M, three similar daughter nuclei of same mass are formed. The speed of a daughter nuclei in terms of mass defect ΔM will be :

(1) $\sqrt{\frac{2c\Delta M}{M}}$

(2) $\frac{\Delta Mc^2}{3}$

(3) $c\sqrt{\frac{2\Delta M}{M}}$

(4) $c\sqrt{\frac{3\Delta M}{M}}$

30) If the energy released in the fission of one nucleus is $3{\cdot}2 \times 10^{-11}$J, then find number of nuclei required per second in a power plant of 16 kW. :-

(1) 5×10^{14}

(2) 5.8×10^{10}

(3) 6×10^{5}

(4) None of these

31)

Light described at a place by the equation
$E=(100\ V/m)[\sin(5\times10^{15}s^{-1})t + \sin(8\times10^{15}s^{-1})t]$ falls on a metal surface having work function 2eV. Calculate the maximum kinetic energy of the photo electrons.

(1) 5.27 eV

(2) 1.71 eV

(3) 3.27 eV

(4) 0.854 eV

32) If a source of power 4kW produces 10^{20} photons/second, the radiation belongs to part of the spectrum called :

(1) γ-rays

(2) X-rays

(3) Ultraviolet rays

(4) Microwaves

33) In an experiment of photoelectric effect when wavelength of incident light is decreased by 60% then KE_{max} of photo electrons increase from 5 eV to 17 eV. The work function of the surface will be :-

(1) 4 eV

(2) 3 eV

(3) 2 eV

(4) 1 eV

34) The de Broglie wavelength of an electron moving with a velocity 1.5×10^8 m/s is equal to that of a photon. The ratio of the kinetic energy of the electron to the energy of the photon is :-

(1) $\frac{1}{4}$

(2) $\frac{1}{2}$

(3) 2

(4) $\frac{4}{1}$

35) An α-particle and a proton are fired through the same magnetic field which is perpendicular to their velocity vectors. The α-particle and the proton move such that radius of curvature of their paths is same. Find the ratio of their de-Broglie wavelengths :-

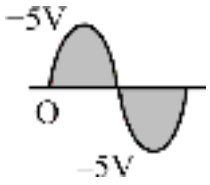
(1) 2 : 3

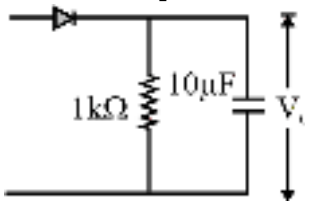
(2) 3 : 4

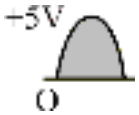
(3) 5 : 7

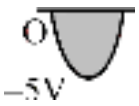
(4) 1 : 2

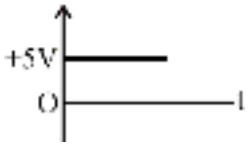
36) The output taken across the capacitor is as shown in figure :

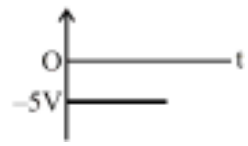




(1) 

(2) 

(3) 

(4) 

37) The number of electron hole pair in a pure silicon crystal at 300 K is 6×10^{15} m^{-3}. When it is

doped by indium the electron concentration becomes 1.2×10^{9} m^{-3}. The number of indium atoms mixed per cubic meter is (approx) :-

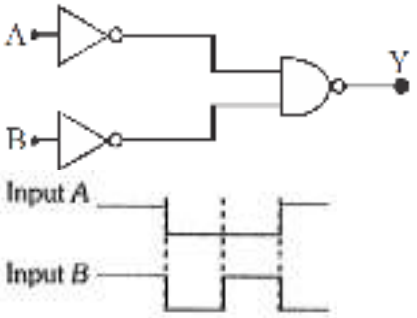
(1) 1.2×10^{9}

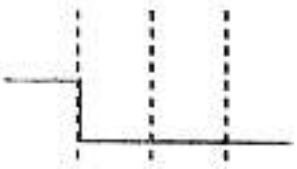
(2) 6×10^{15}

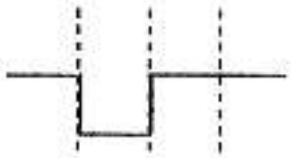
(3) 3×10^{22}

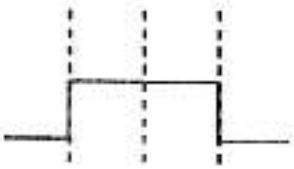
(4) 5×10^{6}

38) The logic circuit shown below has the input waveforms 'A' and 'B' as shown. Pick out the correct output waveform :



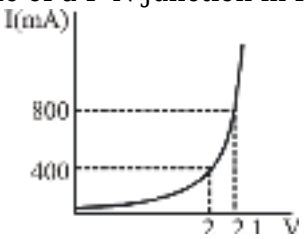
(1) 

(2) 

(3) 

(4) None of these

39) The I-V. characteristic of a P-N junction in forward bias is shown. The approximate forward resistance of junction is-



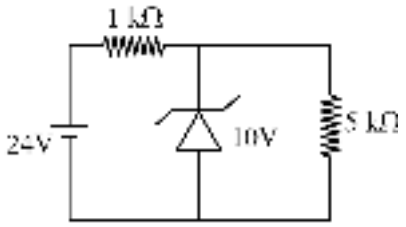
(1) 1Ω

(2) 0.25Ω

(3) 0.5Ω

(4) 5Ω

40) The power produce in zener diode will be :-



(1) 60 mW

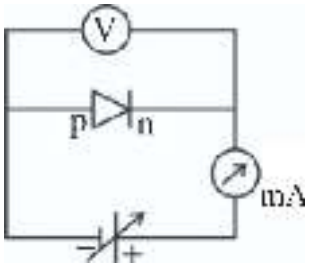
(2) 120 mW

(3) 160 mW

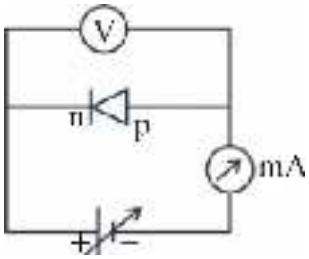
(4) 240 mW

41) Which among the following has non-zero ammeter reading? (voltmeter and ammeter both are ideal)

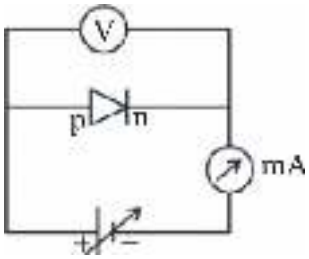
(1)



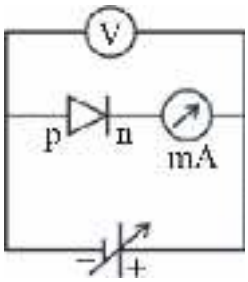
(2)



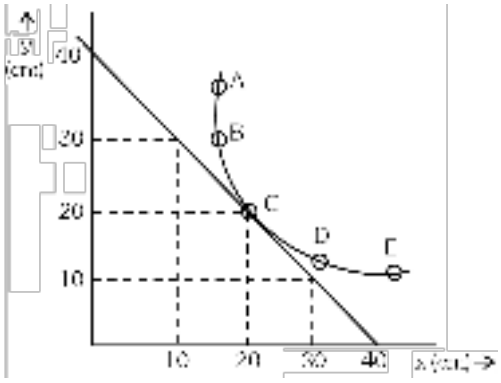
(3)



(4)



42) In an experiment to measure the focal length (f) of a convex lens, the magnitude of object distance (x) and the image distance (y) are measured with reference to the focal point of the lens. The y-x plot is shown in figure.

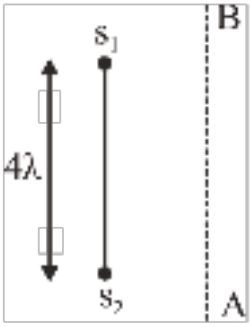
The focal length of the lens is _____ cm. 

(1) 10

(2) 20

(3) 30

(4) 40

43) Two coherent sources each of wavelength λ are separated by a distance 4λ. The total number of minima formed on line AB which runs from –∞ to + ∞ is :- 

(1) 2

(2) 4

(3) 6

(4) 8

44) A radioactive element A converts into element C after emitting 3α and $1\beta^-$. Another element B converts into C by emitting 2α and $5\beta^-$. If mass number and atomic number of A and B are A_1, A_2 and Z_1, Z_2 respectively then which of the following is correct :-

(1) $Z_1 - Z_2 = 5$

(2) $A_1 - A_2 = 4$

(3) both option (1) & (2)

(4) None of these

45) In the network shown in fig. determine V :- 

(1) 11V

(2) 12V

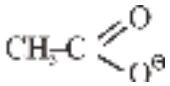
(3) 9.5V

(4) 13V

CHEMISTRY

1) Leaving group tendency is maximum for :-

(1) CH_3–$O^{\ominus}$

(2) 

(3) 

(4) 

2) Among the bromides given below, the order of reactivity of S_N1 reaction is - (I)  (II) 

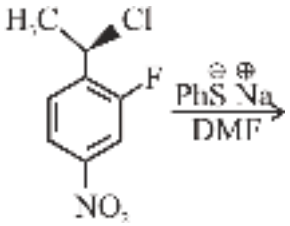
(III) 

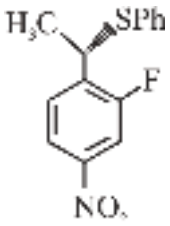
(1) III > I > II

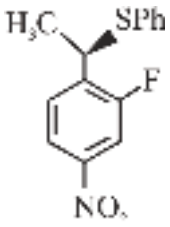
(2) III > II > I

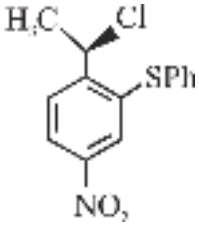
(3) II > III > I

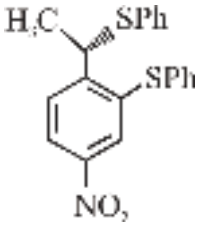
(4) II > I > III

3) The major product of the following reaction is:-  Major product

(1) 

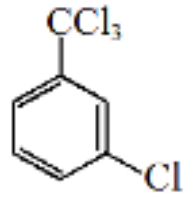
(2) 

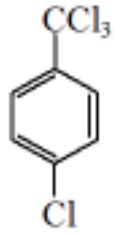
(3) 

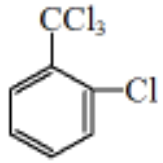
(4) 

4) $C_6H_5CCl_3 \xrightarrow[Fe]{Cl_2}$ X

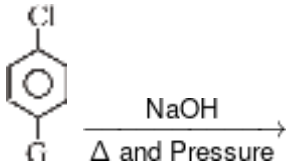
In the above reaction X is

(1) 

(2) 

(3) 

(4) All in equal amount

5) 

Rate of reaction is maximum if G is :-

(1) $-OCH_3$

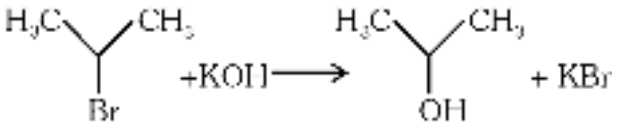
(2) $-CH_3$

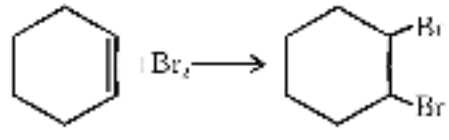
(3) $-NO_2$

(4) –H

6) For the following reactions :-

(a) $CH_3CH_2CH_2Br + KOH \rightarrow CH_3CH{=}CH_2 + KBr + H_2O$

(b) 

(c) 

Which of the following statement is **correct** ?

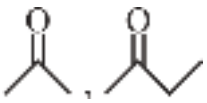
(1) (a) and (b) are elimination reaction and (c) is addition reaction

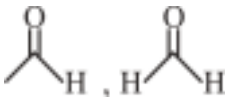
(2) (a) is elimination, (b) is substitution and (c) is addition reaction

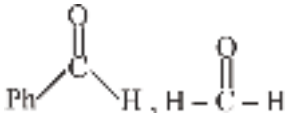
(3) (a) is elimination, (b) and (c) are substitution reactions

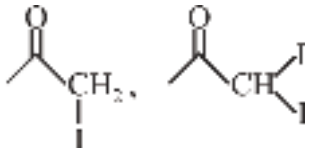
(4) (a) is substitution, (b) and (c) are addition reaction

7) Which of the following pair is differentiated by iodoform test?

(1) 

(2) 

(3) 

(4) 

8) Which is the correct increasing order of boiling points of the following compounds ?
1-iodobutane; 1-bromobutane, 1-chlorobutane and butane

(1) Butane < 1-chlorobutane < 1-bromobutane < 1-iodobutane

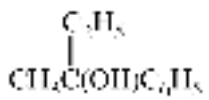
(2) 1-iodobutane < 1-bromobutane < 1-chlorobutane < Butane

(3) Butane < 1-iodobutane < 1-bromobutane < 1-chlorobutane

(4) Butane < 1-chlorobutane < 1-iodobutane < 1-bromobutane

9) $CH_3COOH \xrightarrow{PCl_5} A \xrightarrow[\text{Anhy. } AlCl_3]{C_6H_6} B \xrightarrow[\text{Ether}]{C_2H_5MgBr} C$,
Product 'C' is :-

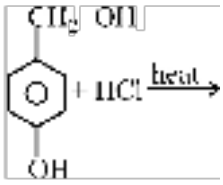
(1) $CH_3CH(OH)C_6H_5$

(2) 

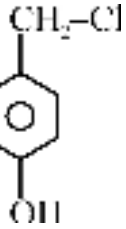
(3) $CH_3CH(OH)C_2H_5$

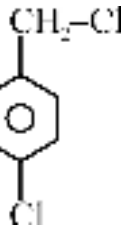
(4) $CH_3COC_6H_5$

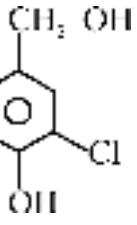
10) The major product in the following reaction –

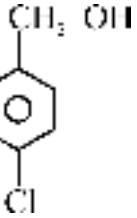


is :-

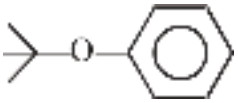
(1) 

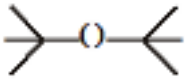
(2) 

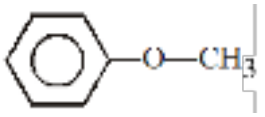
(3) 

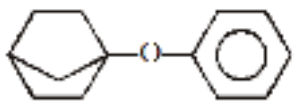
(4) 

11) Which of the following ether can be synthesized through Williamson ether synthesis ?

(1) 

(2) 

(3) 

(4) 

12) $CH_3–CH_2–CH_2–OH \xrightarrow[\text{reagent}]{\text{Jone's}} (P) \xrightarrow{PCl_5} (Q) \xrightarrow{H_2+Pd+BaSO_4} (R)$

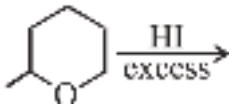
Product P, Q and R respectively.

(1) CH_3CH_2COOH, CH_3CH_2COCl, $CH_3CH_2CH_2Cl$

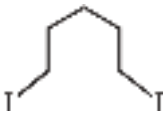
(2) CH_3CH_2COOH, CH_3CH_2CHO, CH_3CH_2COCl

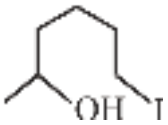
(3) CH_3CH_2COOH, CH_3CH_2COCl, CH_3CH_2CHO

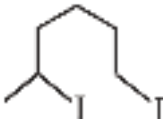
(4) CH_3COOH, CH_3COCl, CH_3CHO

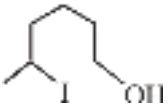
13) 

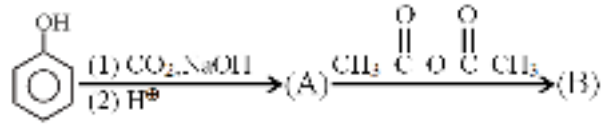
Product is :-

(1) 

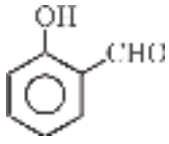
(2) 

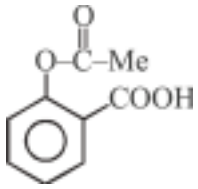
(3) 

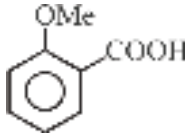
(4) 

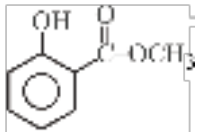
14) 

Product (B) is :

(1) 

(2) 

(3) 

(4) 

15) **Statement-I :** Tertiary alcohol produce turbidity immediately with Lucas reagent.
Statement-II : Reaction of primary alcohol with HCl require the pressure of a catalyst $ZnCl_2$.

(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

(3) Statement I is true but Statement II is false.

(4) Statement I is false but Statement II is true.

16) **Assertion (A) :** Ethanol has higher boiling point than that of dimethyl ether.
Reason (R) : Molecules of ethanol are associated through intermolecular H–bonding while in dimethyl ether there is no intermolecular hydrogen bonding with each other.

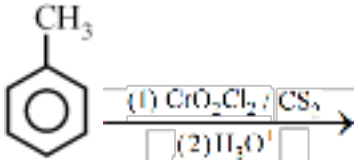
(1) Both assertion and reason are correct and reason is the correct explanation of assertion

(2) Both assertion and reason are incorrect

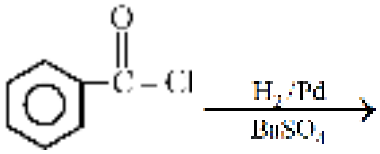
(3) Assertion is correct and reason is incorrect

(4) Both assertion and reason are correct and reason is not the correct explanation of assertion

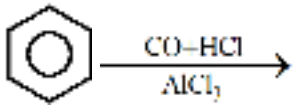
17) Which of the following reaction name is correct.

(1) 

Etard reaction

(2) 

Gatterman koch reaction

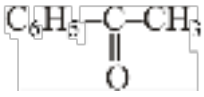
(3) 

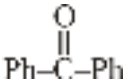
Kolbe reaction

(4) CH_3–CN $\xrightarrow[(2)\ H_3O^+]{(1)\ SnCl_2-HCl}$
Resenmund reaction

18) Which compound is least reactive for nucleophilic addition reaction :-

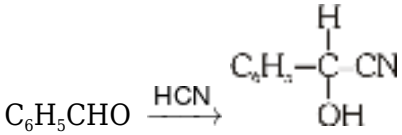
(1) C_6H_5CHO

(2) 

(3) 

(4) 

19)



The product would be

(1) a meso compound

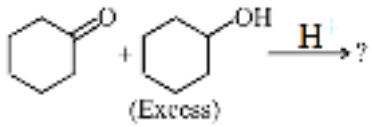
(2) Only d

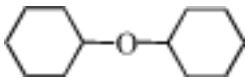
(3) Only □

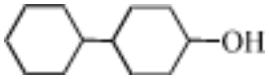
(4) Racemic mixture

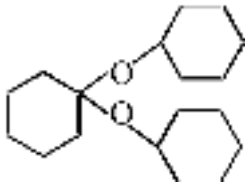
20)

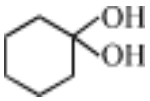
What is the major product of the following reaction:-



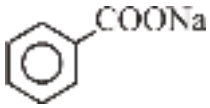
(1) 

(2) 

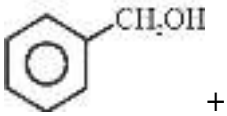
(3) 

(4) 

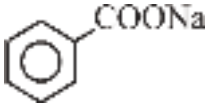
21) Reaction of Benzaldehyde with formaldehyde in the presence of 50% conc. NaOH mainly gives :-

(1) 

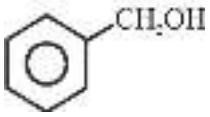
+ CH_3OH

(2) 

+ CH_3CH_2ONa

(3) 

+HCOONa

(4) HCOONa + 

22) $CH_3CHO + CH_3CH_2CHO \xrightarrow[\Delta]{OH^{\ominus}}$ Products, Which one is not obtained during this reaction :

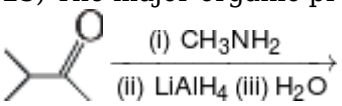
(1) Pent-2-enal

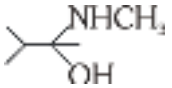
(2) 2-Methylbut-2-enal

(3) But-2-enal

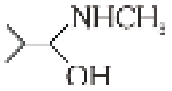
(4) 3-Methylpent-2-enal

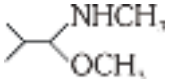
23) The major organic product formed from the following reaction is :–



(1) 

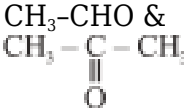
(2) 

(3) 

(4) 

24)

Which of the following is not correctly matched?

	Compound	Test of differentiation
(1)	CH_3CHO & 	Fehling's Test
(2)	$CH_3–C\equiv C–CH_3$ & $CH_3–CH_2–C\equiv CH$	Tollen's reagent
(3)	$CH_3–CHO$ & 	Iodoform Test
(4)	$CH_3–CHO$ & $Ph–CHO$	Fehling's Test

(1) 1

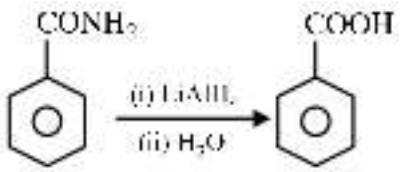
(2) 2

(3) 3

(4) 4

25) Select the **incorrect** reaction among the following :

(1) $CH_3COCl \xrightarrow[\Delta]{H_2O} CH_3COOH$

(2) 

(3) $CH_3CH_2OH \xrightarrow[(ii)\,H_3O^{\oplus}]{(i)\,KMnO_4/OH^-} CH_3COOH$

(4) $CH_3CH_2CH_2OH \xrightarrow{CrO_3–H_2SO_4} CH_3CH_2COOH$

26) The reaction of ethanol on acetic anhydride is an example of :-

(1) Nucleophilic addition
(2) Nucleophilic substitution
(3) Electrophilic addition
(4) Free radical substitution

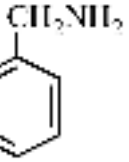
27) When chlorine is treated with carboxylic acid in the presence of red phosphorus then α-chloro acid will be formed, this reaction is known as :

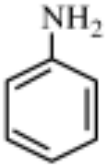
(1) Hunsdicker reaction
(2) Hell-Volhard - Zelinsky reaction
(3) Friedel - crafts reaction
(4) Rosenmund reduction

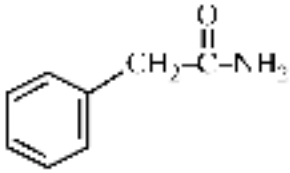
28) $Ph–C\equiv N \xrightarrow[\text{(Partial hydrolysis)}]{\overset{\ominus}{O}H/H_2O} (A) \xrightarrow{Br_2+KOH} (B)$,
Product (B) is :-

(1) $Ph – CH_2 – NH_2$
(2) $Ph – OH$
(3) $Ph – NH_2$
(4) $Ph – CH_3$

29) Which of the following compounds can be prepared in good yield by Gabriel phthalimide synthesis?

(1) 

(2) 

(3) 

(4) $CH_3–CH_2–NHCH_3$

30) Aniline on treatment with bromine water yields white precipitate of :-

(1) o–Bromoaniline
(2) p–Bromonailine

(3) 2, 4, 6–Tribromoaniline

(4) m–Bromoaniline

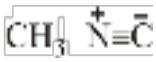
31)

$CH_3NH_2 + CHCl_3 + KOH \rightarrow (x) + KCl + H_2O$,
x is :-

(1) $CH_3–C\equiv N$

(2) $CH_3–NH–CH_3$

(3) 

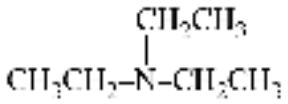
(4) 

32) An organic compound react with aq. HNO_2 to produce an oily nitrosoamine. The compound is :-

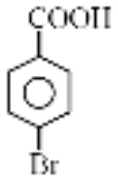
(1) $CH_3–NH_2$

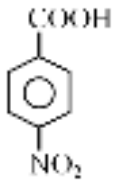
(2) $CH_3CH_2\ NH_2$

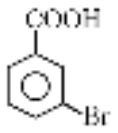
(3) $CH_3CH_2NHCH_2CH_3$

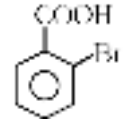
(4) 

33) 
E is :-

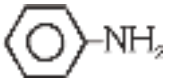
(1) 

(2) 

(3) 

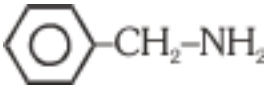
(4) 

34) Which compound does not show diazo coupling reaction:-

(1) 

(2) 

(3) 

(4) 

35) Match the column

	Column-I Vitamin		**Column-II Deficiency diseases**
(A)	Vitamin A	(i)	Scurvy
(B)	Vitamin C	(ii)	Rickets
(C)	Vitamin B_1	(iii)	Xerophthalmia
(D)	Vitamin D	(iv)	Beri-Beri

(1) A-iii, B-i, C-iv, D-ii

(2) A-iii, B-i, C-ii, D-iv

(3) A-i, B-iii, C-ii, D-iv

(4) A-iii, B-iv, C-i, D-ii

36) Which of the following statement is correct.

(1) Proteins are most abundant biomolecules of the living system

(2) Cellulose is most abundant organic substance in plant kingdom

(3) Glucose is most abundant organic compound on earth

(4) All of the above

37) α–D –(+) glucose and β-D-(+) glucose are :-

(1) epimers

(2) anomers

(3) enantiomers

(4) conformers

38) **Statement-I :** The pentaacetate of glucose does not react with H_2N-OH indicating the absence of free –CHO group.
Statement-II : Glucose is found to exist in two different crystalline form which was named as α and β glucose.

(1) Both Statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct and statement-II is incorrect

(4) Statement-I is incorrect and statement-II is correct

39) Which of the following hormones contains iodine ?

(1) Thyroxine
(2) Insulin
(3) Testosterone
(4) Adrenaline

40) Naturally occuring (+) sucrose is :

(1) α–D glucopyranoside +β–D– fructofuranoside
(2) α–D glucopyranoside +α–D– fructofuranoside
(3) β–D glucopyranoside +α–D– fructofuranoside
(4) β–D glucopyranoside +β–D– fructofuranoside

41) Which of the following does not give white precipitate when boiled with alcoholic silver nitrate?

(1) Allyl chloride
(2) t-butyl chloride
(3) Chlorobenzene
(4) Benzyl chloride

42) In a Lassaignes's test for sulphur in the organic compound with sodium nitroprusside solution the purple colour formed is due to-

(1) $[Fe(CN)_5\ NOS]^{4-}$
(2) $[Fe(CN)_5\ S]^{2-}$
(3) $[Fe(CN)_5\ NOS]^{2-}$
(4) $[Fe(CN)_6\ ]^{4-}$

43) While preparing lassaigne solution nitrogen containing organic compound when fused with sodium metal forms ?

(1) $NaNO_2$
(2) NaCN
(3) $NaNH_2$
(4) NaNC

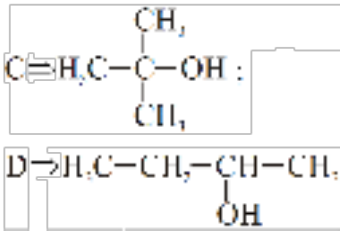
44) Phenol does not react with which of the following?

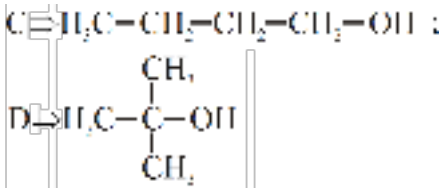
(1) Alkali metals
(2) Sodium hydroxide
(3) Potassium hydroxide
(4) Sodium bicarbonate

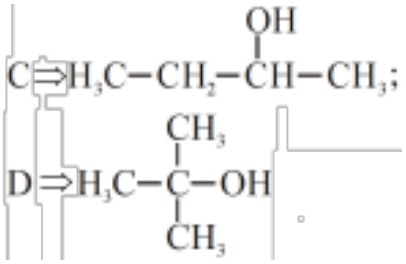
45) Two compounds A and B with same molecular formula C_3H_6O undergo Grignard's reaction with methylmagnesium bromide followed by acidic hydrolysis to give products C and D. Products C and D show following chemical tests.

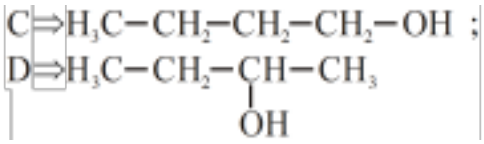
Test	C	D
Ceric ammonium nitrate Test	Positive	Positive
Lucas Test	Turbidity obtained after five minutes	Turbidity obtained immediately
Iodoform Test	Positive	Negative

C and D respectively are :

(1) 

(2) 

(3) 

(4) 

BIOLOGY

1) The wasp pollinates the fig inflorescence while searching for suitable.....sites.

(1) Pollination

(2) Oviposition

(3) Pseudocopulation

(4) Co-Evolution

2) Clown fish get protection from predators in the presence of sea anemone because they have :

(1) Large sized body

(2) Pointed teeth

(3) Stinging tentacle

(4) Haustoria

3) The..... depend on two intermediate hosts (a snail and a fish) to complete its life cycle.

(1) Human liver fluke

(2) Leech

(3) Copepods

(4) Ticks

4) Which of the following is an example of endoparasite ?

(1) Leech on cattle

(2) Ticks on dogs

(3) Lice on human

(4) *Plasmodium*

5) The prickly pear cactus introduced into Australia in the early.....caused havoc by spreading rapidly into millions of hectares of rangeland.

(1) 1930

(2) 1920

(3) 1940

(4) 1950

6) In this interaction one species is inhibited by toxic secretion of other species :

(1) Amensalism

(2) Competition

(3) Predation

(4) Parasitism

7) Given below are two statements :
Statement – I : In many species of fig there is tight one to one relationship with the pollinator species of wasp.
Statement – II : It means that a given fig species can be pollinated only by it's partner wasp species and no other species.
In the light of the above statements choose the correct answers from the options given below.

(1) Both statement I and statement II are false

(2) Statement I is correct but statement II is false

(3) Statement I is incorrect but statement II is true

(4) Both statement I and statement II are true

8) Read the following Assertion and Reason and choose the correct option.

Assertion : The activity of key-stone species determines the structure of the community.
Reason: Key stone species are present in low abundance.

(1) Both Assertion and Reason are true but Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is correct but Reason is incorrect.
(4) Assertion is incorrect but Reason is correct.

9) Identify the incorrect statement-

(1) Darwin spoke of the struggle for existence and survival of the fittest in nature.
(2) Interspecific competition is a potent force in organic evolution.
(3) Totally unrelated species could also compete for the same resource.
(4) In shallow south American lakes, visiting flaminogoes and resident fishes compete for phytoplankton

10) Which of the following statement is incorrect regarding predation?

(1) It keeps prey population under control.
(2) It helps in maintaining species diversity in a community.
(3) It increases the intensity of competition among prey species.
(4) Predators are prudent in nature

11) Read the following column-I & column-II and identify correct match :–

	Column-I		**Column-II**
(A)	Mac Arther	(i)	Resource partitioning
(B)	Intraspecific competition	(ii)	Gause
(C)	Scavenger	(iii)	David Tilman
(D)	Competitive exclusion principle	(iv)	More acute
		(v)	Vulture

(1) (A)-(i), (B)-(ii), (C)-(iii), (D)-(iv)
(2) (A)-(ii), (B)-(iii), (C)-(v), (D)-(i)
(3) (A)-(i), (B)-(iv), (C)-(v), (D)-(ii)
(4) (A)-(iv), (B)-(iii), (C)-(i), (D)-(v)

12) **Assertion (A)** : Starfish Pisaster is considered a keystone species in the intertidal communities of the American Pacific Coast.
Reason (R) : The removal of Pisaster led to the extinction of several species due to increased interspecific competition.

(1) Both A and R are true, and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

(3) A is true, but R is false.

(4) A is false, but R is true.

13) Read the following statements and select the correct option :
(i) Parasites acting as conduits for energy transfer across trophic levels.
(ii) Competition is a process in which the fitness of one species is significantly lower in the presence of another species.
(iii) The human liver fluke depends on two intermediate hosts to complete its life cycle.
(iv) Lichens represent an intimate mutualistic relationship between a fungi and the roots of higher plants.

(1) Only (ii) and (iii) are correct.

(2) Only (i) is correct.

(3) (i), (ii), (iii) and (iv) are correct.

(4) Only (ii), (iii) and (iv) are correct.

14) (i) ...A... refers to the number of births during a given period in the population that are added to the initial density.
(ii) ...B... is the number of deaths in the population during a given period.
(iii) ...C... is the number of individuals of the same species that have come into the habitat from elsewhere during the time period under consideration.
(iv) ...D... is the number of individuals of the population who left the habitat and gone elsewhere during the time period under the consideration.
Fill in the blanks with appropriate option ?

(1) A-Emigration ; B-Immigration
C-Natality ; D-Mortality

(2) A-Natality ; B-Mortality
C-Immigration ; D-Emigration

(3) A-Emigration ; B-Mortality
C-Natality ; D-Immigration

(4) A-Natality ; B-Mortality
C-Emigration ; D-Immigration

15) Match the column I with column II.

Column-I		**Column-II**	
(a)	Higher number of pre reproductive, moderate number of reproductive and fewer post reproductive individuals.	(i)	Declining population
(b)	An equal number of pre reproductive and reproductive individuals	(ii)	Expanding population
(c)	Fewer number of pre reproductive compared to reproductive individual	(iii)	Stable population

(1) a–i ; b–ii, c–iii

(2) a–iii ; b–ii, c–i

(3) a–ii ; b–iii, c–i
(4) a–ii ; b–i, c–iii

16) Of the total incident solar radiation Photosynthetic active radiation (PAR) is :-

(1) Less than 50%
(2) More than 50%
(3) 2-10%
(4) Equal to 50%

17) What is the main function of a detritivores in an ecosystem :

(1) Pollination
(2) Parasitism
(3) Pathogenic
(4) Nutrient recycling

18) Which of the following food chains transfers the largest fraction of energy in a terrestrial ecosystem :-

(1) DFC
(2) GFC
(3) Parasitic food chain
(4) All of the above

19) The pyramid of biomass in sea is generally :-

(1) Upright
(2) Inverted
(3) Spindle shape
(4) 1 and 2 both

20) Which of the following are called as key industry animals.

(1) Autotrophs
(2) Decomposers
(3) Herbivores
(4) Top carnivores

21) Earth is _____ system for energy.

(1) Open
(2) Closed
(3) Both (1) and (2)
(4) Cyclic

22) Choose the incorrect statement :-

(1) All animals depend on producers directly or indirectly for their food needs.

(2) Primary consumer will be herbivores.

(3) The consumer that feed on herbivores are carnivores.

(4) Animals that depend on the herbivores for food are called secondary carnivores.

23) Match List-I with List-II

	List-I		**List-II**
(A)	Fragmentation	(I)	Breakdown of detritus in to smaller particles.
(B)	Catabolism	(II)	Release of inorganic nutrients
(C)	Mineralisation	(III)	Degrade detritus into simpler inorganic substances.

Choose the correct answer from the options given below :-

(1) A-I, B-III, C-II

(2) A-I, B-II, C-III

(3) A-II, B-III, C-I

(4) A-III, B-I, C-II

24) What is relationship between the standing crop and productivity of an ecosystem.

(1) Standing crop is the amount of organic matter produced per unit area, while productivity is the total amount of organic matter present in an ecosystem.

(2) Productivity is the amount of organic matter produced per unit area per unit time while standing crop is the total amount of living organic matter present in ecosystem.

(3) Standing crop and productivity are unrelated concepts in ecology.

(4) Standing crop and productivity is the amount of organic matter produced per unit area.

25) Read statements carefully :

Statement-I : Identification and enumeration of plant and animal species of an ecosystem gives its species composition.

Statement-II : The food chain or web is formed because of the interdependency in ecosystem.

(1) **Statements I & II** both are correct

(2) **Statements I & II** both are incorrect

(3) **Statement-I** is correct only

(4) **Statement-II** is incorrect only

26) Select incorrect statements :

(1) Water is the abiotic component with all the dissolved inorganic and organic substances and the rich soil deposit at the bottom of the pond ecosystem.

(2) Tree occupy top vertical strata or layer of a forest.

(3) All Animals depends on producers for their food needs.

(4) The energy trapped by the producer is either not passed on to a consumer or the organism dies.

27) What role do macro-consumers typically play in ecosystem :

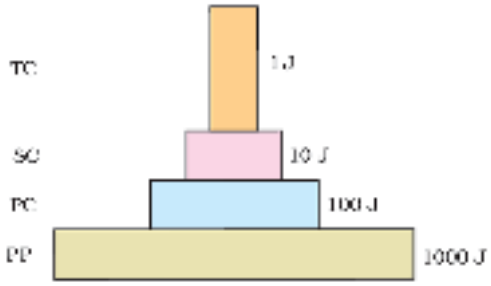
(1) They regulate population sizes of primary producers

(2) They are responsible for photosynthesis

(3) They fix N_2 in soil

(4) They decompose dead organic matter and recycle nutrients

28)



The above is an ideal pyramid of energy of an ecosystem. What is the total amount of sunlight coming into the ecosystem :-

(1) 1,000,000 J

(2) 1,00,000 J

(3) 1000 J

(4) 100 J

29) Read the given following statements :-
(a) Estuary is not an example of aquatic ecosystems
(b) The animal who eat other herbivore animals are called primary consumers
(c) Some common herbivores are insects, birds and mammals in terrestrial ecosystem.
(d) Obviously the secondary consumers will be herbivores
(e) Ecosystem not varies greatly in size
Choose the correct answer from the options given below :-

(1) (b), (d), (e) only

(2) (a), (b), (c), (d), (e)

(3) (a), (c), (d), (e) Only

(4) (c) only

30) Read the following statements -
(i) Detritivores break down detritus into smaller particles called fragmentation.
(ii) Bacterial and Fungi enzymes degrade detritus into simpler inorganic substances called catabolism.
(iii) The green plant in the ecosystem are called producers.
(iv) In an aquatic ecosystem DFC is the major conduit for energy flow.
Which of the above statements are correct.

(1) (i) Only

(2) (iii) and (iv) Only

(3) (i), (ii) and (iii) only

(4) All of the above

31) Which level of biodiversity refers to the diversity within a single species ?

(1) Genetic diversity

(2) Species diversity

(3) Ecological diversity

(4) Ecosystem diversity

32) The number of _____ species in the world is more than the combined total of the species of fishes, amphibians, reptiles and mammals.

(1) Algae

(2) Lichens

(3) Fungi

(4) Ferns

33) India, with much of it's land area in the tropical latitudes has _____ species of birds. Complete the statement using most appropriate option among the following options.

(1) less than 400

(2) more than 1400

(3) more than 1200

(4) only 3000

34) _____showed in his experiments, that increased diversity contributed to higher productivity.

(1) Paul Ehrlich

(2) Robert May

(3) David Tilman

(4) Reiter

35) How many birds species are facing threat of extinction in the world ?

(1) 23 percent

(2) 12 percent

(3) 32 percent

(4) 31 percent

36) How many species are example of Alien species invasions ?
Nile perch, lantana, water hyacinth, parthenium, African catfish, passenger pigeon.

(1) 4

(2) 5

(3) 6

(4) 3

37) **Assertion (A) :-** Economic benefits from nature are placed in narrow utilitarian.
Reason (R) :- Exploring molecular, genetic and species-level diversity for products of economic importance is known as bio prospecting.

(1) Both Assertion and Reason are correct but Reason is the correct explanation of Assertion

(2) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion

(3) Assertion is correct but Reason is incorrect

(4) Assertion is not correct but Reason is correct

38) **Statement-I :** Biodiversity hotspots are regions with very high levels of species richness & high degree of endemism.
Statement-II : There are many animals that have become extinct in the wild but continue to be maintained in zoological parks.

(1) Both statements are incorrect

(2) Statement-I is correct and Statement-II is incorrect

(3) Statement-I is incorrect and Statement-II is correct

(4) Both Statements are correct

39) What was the main objective of the convention on Biological Diversity held in 1992 ?

(1) To promote industrial development

(2) To encourage deforestation

(3) To call upon nations for biodiversity conservation and sustainable utilization of its benefits

(4) To discuss political issues

40) Match the column-I with column-II :-

	Column-I		**Column-II**
(A)	Extreme high risk of extinction in immediate future	(i)	Endangered
(B)	Very high risk of extinction in wild in near future	(ii)	Vulnerable
(C)	High risk of extinction in medium term future	(iii)	Critically endangered

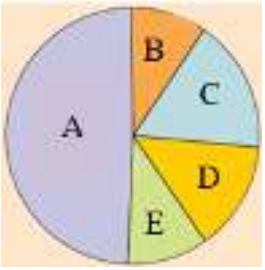
(1) A-i, B-ii, C-iii

(2) A-iii, B-ii, C-i

(3) A-iii, B-i, C-ii

(4) A-ii, B-iii, C-i

41) Global diversity of vertebrate is represented in the image. Choose correct combination-



	A	B	C	D	E
(1)	Fishes	Bird	Reptiles	Amphibia	Mammals
(2)	Fishes	Mammals	Bird	Reptile	Amphibia
(3)	Fishes	Reptile	Bird	Mammals	Amphibia
(4)	Fishes	Mammals	Reptiles	Bird	Amphibia

(1) 1

(2) 2

(3) 3

(4) 4

42) Identify the incorrect statement :

(1) According to IUCN the total number of plant and animal species are more than 1.5 million

(2) For many taxonomic groups, species inventories are more complete in temperate than in tropical countries

(3) According to Robert May global species diversity at about 7 million

(4) More than 70% of all the species recorded are plant

43) Below two statement are given.
Statement-I : The IUCN Red List (2004) document the extinction of 784 species in the last 500 years.
Statement-II : The accelerated rates of species extinction that the world is facing now are largely due to human activities.

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct and Statement-II is incorrect.

(3) Statement-I is incorrect and Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

44) Match column-I with column-II (benefits) and select the correct option :

Column-I		Column-II	
(1)	Broadly utilitarian	(A)	Nature is the main resource of food, wood, fabrics and resins

(2)	Narrowly utilitarian	(B)	Every species in nature has an intrinsic value
(3)	Ethical reason	(C)	Plant cover protects soil from wind and water erosion

(1) 1-B, 2-A, 3-C

(2) 1-A, 2-C, 3-B

(3) 1-C, 2-A, 3-B

(4) 1-B, 2-C, 3-A

45) **Assertion (A) :** Tropical latitudes had a long evolutionary time for species diversification.
Reason (R) : Unlike temperate region, tropical latitudes have remained relatively undisturbed for millions of years.

(1) Both A and R are true and R is correct explanation of A.

(2) Both A and R are true but R is not correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

46) Which of the following statement is false ?

(1) Active immunity develops after a lag period

(2) Active immunity is very effective in immunodeficient host

(3) Passive immunity is used for the treatment of Acute infection

(4) No immunological memory is formed by passive immunity

47) Match the column-I & column-II & choose the option with correct Matching.

	Column-I		**Column-II**
(1)	Pneumonia	(a)	*Plasmodium*
(2)	Typhoid	(b)	*Salmonella typhi*
(3)	Common cold	(c)	*Haemophilus influenzae*
(4)	Malaria	(d)	Rhino virus

(1) 1-c, 2-a, 3-b, 4-d

(2) 1-d, 2-b, 3-a, 4-c

(3) 1-c, 2-b, 3-d, 4-a

(4) 1-a, 2-d, 3-b, 4-c

48) **Assertion :-** Metastasis is the most feared property of malignant tumors.
Reason :- Cells sloughed from such tumors reach distant sites through blood and interfere in functions of other tisues.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

49) Choose the option which describes the physical barrier of innate immunity :-

(1) Acid in the stomach, saliva in the mouth, tears from eyes prevent microbial growth

(2) Mucus coating of the epithelium lining the respiratory, gastrointestinal and urinogenital tracts help in trapping microbes entering our body

(3) Chemical like interferons protects cells from viral infection

(4) Rejection of an allograft

50) Select the incorrect statement against Malaria.

(1) Sexual stage of plasmodium develops in human.

(2) Fertilization of plasmodium takes place in gut of mosquito.

(3) Plasmodium parasite reproduce sexually in liver cell.

(4) Plasmodium is transmitted through female anopheles vector.

51) **Statement-I :** Health for a long time was considered as a state of body and mind where there was a balance of certain 'humors'.
Statement-II : It was thought that the persons with 'black bile' belonged to hot personality and would have fever.

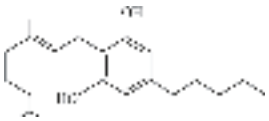
(1) Statement-I and Statement-II both are correct.

(2) Statement-I and Statement-II both are incorrect.

(3) Only Statement-I is correct.

(4) Only Statement-II is correct.

52) Which pair is matched correctly ?

(1)		Flowering branch of Datura
(2)		Chemical structure of morphine
(3)		Leaves of *Cannabis sativa*
(4)		Opium poppy

(1) 1

(2) 2

(3) 3

(4) 4

53) _____ an intestinal parasite causes blockage of the intestinal passage, internal bleeding, muscular pain, anaemia and fever.

(1) *Entamoeba histolytica*

(2) Filarial worm

(3) *Ascaris*

(4) Typhoid

54) Where immature lymphocytes differentiate into antigen sensitive lymphocytes :-

(1) Spleen

(2) Lymph nodes

(3) Both (1) & (2)

(4) Bone marrow

55) Person who take drugs intravenously is much more likely to acquire serious infections like ?

(1) Liver cirrhosis

(2) Cancer

(3) Hepatitis-B

(4) Dependence

56) Which of the following stimulates the secretion of adrenaline ?

(1) Cocaine

(2) LSD

(3) Smack

(4) Tobacco

57)

Consider the following four statements (A–D) and select the option which includes all the correct ones only :-
(A) All parasites are pathogens as they do not cause harm to the host by living in (or on) them.
(B) Balanced diet, personal hygiene and regular exercise are very important to maintain good health.
(C) Health also increases longevity of people and reduces infant and maternal mortality.
(D) Health simply means 'absence of disease' or 'physical fitness'.

(1) A, B, C

(2) B and C only

(3) B, C, D

(4) A, B, C, D

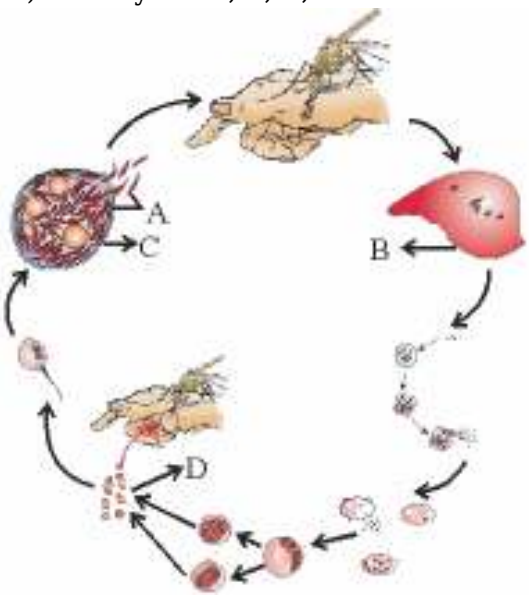
58) **Assertion (A) :-** Ascaris, an intestinal parasite, causes ascariasis which includes symptoms such as internal bleeding, muscular pain, fever, anemia, and blockage of the intestinal passage.
Reason (R) :- The eggs of the parasite are excreted along with the faeces of infected persons which contaminate soil, water, plants etc. A healthy person acquires this infection through contaminated water, vegetables, fruits, etc.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(3) **Assertion** is true but **Reason** is false.
(4) **Assertion** is false but **Reason** is true.

59) **Statement-I :** Because of the addictive nature of alcohol and drug and their perceived benefit like relief from stress, a person may try taking these in the face of peer pressure, examination-related stress.
Statement-II : Some of us are sensitive to some particle in environment and suddenly started sneezing because of allergy to mites etc.

(1) Statement-I is correct and II is incorrect
(2) Statement-II is correct and I is incorrect
(3) Statement-I & II both are correct.
(4) Statement-I & II both are incorrect.

60) Identify the A, B, C, D in the following life cycle of plasmodium.



Options :-

	A	B	C	D
(1)	Gametocyte	Salivary gland	Liver	Sporozoites
(2)	Sporozoites	Salivary gland	Liver	Gametocyte
(3)	Sporozoites	Liver	Salivary gland	Gametocyte
(4)	Sporozoites	Liver	Gametocyte	Salivary gland

(1) 1

(2) 2

(3) 3

(4) 4

61) The immediate adverse effects of drugs is/are?

(A) Anxiety (B) Nausea
(C) Vandalism (D) Violence
(E) Shakiness (F) Reckless behaviour
(G) Sweating

(1) B, C, E, G

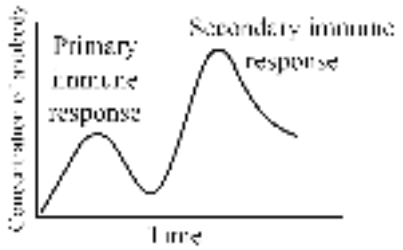
(2) A, B, E, G

(3) C, D, F

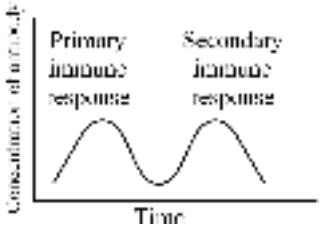
(4) A, B, C, D, E, F, G

62) Which of the following graph is correct for aquired immunity ?

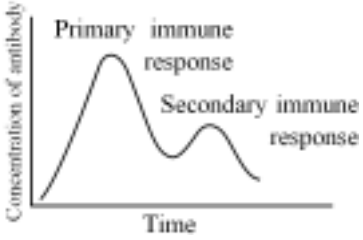
(1)



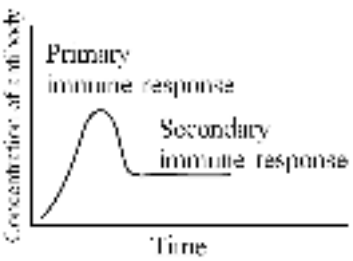
(2)



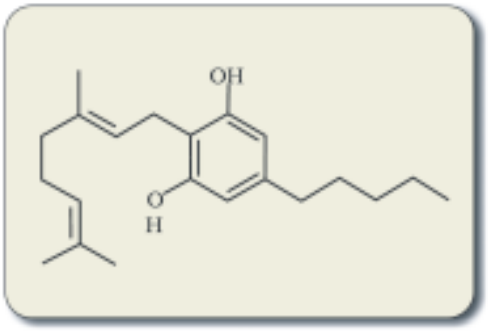
(3)



(4)



63) Identify the molecule shown below and select the right option :-



(1) Morphine
(2) Cocain
(3) Cannabinoid
(4) LSD

64) Which feature of the acquired immunity is the basis of secondary or anamnestic immune response?

(1) Specificity
(2) Diversity
(3) Discrimination between self and non-self cells
(4) Memory

65) Hashish and Ganja are got from :

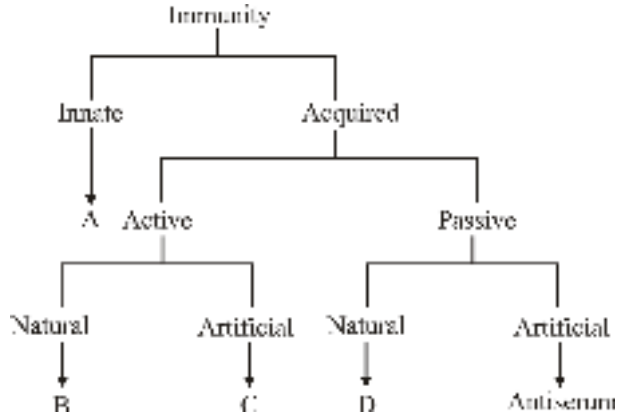
(1) Erythroxylum
(2) Heroin
(3) Papaver
(4) Cannabis

66) Find the correct statement about chemical which is obtained from given plant.



(1) Excessive doses causes hallucination
(2) Abused by some sport person
(3) Producing a sense of euphoria
(4) These are known for their effects on cardiovascular system of the body

67) Study the given flow chart and identify A, B, C, D :-



	A	B	C	D
(1)	Infection	Interferon	Vaccine	Antiserum
(2)	Interferon	Infection	Antiserum	Vaccine
(3)	Infection	Vaccine	Placenta	Interferon
(4)	Interferon	Infection	Vaccine	Placenta (IgG)

(1) 1

(2) 2

(3) 3

(4) 4

68)

Match the following column A column B.

Column-A		Column-B	
(i)	Cytokine barrier	(a)	Mucous coating of epithelium
(ii)	Physical barrier	(b)	Interferons
(iii)	Physiological barrier	(c)	Tears from eyes
(iv)	Cellular barrier	(d)	PMNL-Neutrophils

Choose the correct option from below.

	(i)	(ii)	(iii)	(iv)
(1)	a	d	b	c
(2)	a	c	b	d
(3)	b	a	c	d
(4)	b	a	d	c

(1) 1

(2) 2

(3) 3

(4) 4

69) Which of the following statement is false ?

(1) The adverse effects are just not restricted to the person who is using drug or alcohol.

(2) At times, a drug/alcohol addict becomes the cause of mental and financial distress to his/her entire family and friends.

(3) Excessive doses of drug may lead to coma and death due to respiratory failure, heart failure or cerebral haemorrhage.

(4) Withdrawal symptoms can never be severe or life threatening.

70) Which of the following is used in the production of the recombinant Hepatitis B vaccine ?

(1) Yeast

(2) Microsporum

(3) E.coli

(4) Streptococcus

71) The receptors for opioids are found in :-

(1) GIT & CNS

(2) CNS & CVS

(3) CVS & GIT

(4) GIT and Urinary system

72) Smoking increases _____ content in blood and reduces the concentration of haembound oxygen.

(1) CO_2

(2) CO

(3) O_2

(4) CH_4

73) The exaggerated response of the immune system to certain antigens present in the environment could not be treated by :-

(1) Adrenaline

(2) Steroids

(3) (1) and (2) both

(4) Histamine

74) Which statement is **correct** for industrial melanism?

(1) It is a directional natural selection where median population is selected.

(2) It is a directional natural selection where either of the two extremities are selected at particular time.

(3) It is a stabilising natural selection where median population is selected.

(4) It is a stabilising natural selection where one of the extremities is selected.

75) Identify the correct match from column I and column II :–

Column–I	Column–II
(A) Origin of Universe	(i) 4 bya
(B) Origin of earth	(ii) 4.5 bya
(C) Origin of first cellular form of life	(iii) 3 bya
(D) Origin of first non-cellular form of life	(iv) 13.8 bya
(E) Life appeared	(v) 2 bya

(1) A – iv, B – i, C – iii, D – ii, E – v

(2) A – iv, B – i, C – v, D – ii, E – iii

(3) A – iv, B – ii, C – v, D – iii, E – i

(4) A – iv, B – ii, C – iii, D – v, E – i

76) **Assertion :-** The thorn of Bougainvillea and tendrils of cucurbita represent homology.
Reason :- Same structure developed in different direction according to their need of adaptation.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

77) **Statement-I :** Evolution is not a directed process in the sense of determinism.
Statement-II : Evolution is a stochastic process based on chance events in nature and chance mutation in the organisms.

(1) Both Statement-I and Statement-II are incorrect

(2) Statement-I is correct but Statement-II is incorrect

(3) Statement-I incorrect and Statement-II is correct

(4) Both Statement-I and Statement-II are correct

78) Which of the following is the **correct** sequence for evolution of plants ?

(1) Rhynia-type plants → Psilophyton → Gnetales.

(2) Psilophyton → Progymnosperms → seed ferns → Ferns.

(3) Rhynia-type plants → Psilophyton → Herbaceous lycopods.

(4) Psilophyton → Seed ferns → Cycads → Angiosperms.

79) In which condition the gene ratio remains constant for any species?

(1) Sexual selection
(2) Random mating
(3) Mutation
(4) Gene flow

80) Who has the lowest cranial capacity ?

(1) *Homo erectus*
(2) *Homo habilis*
(3) Neanderthal man
(4) *Homo sapiens sapiens*

81) Life cannot originate from inorganic materials now because of :

(1) High level of CO_2
(2) Low temperature
(3) High atmospheric oxygen
(4) Absence of oxygen

82) Fishes with stout and strong fins could move on land and go back to water occurred about ?

(1) 350 mya
(2) 200 mya
(3) 65 mya
(4) 250 mya

83) Darwin finches of Galapagos islands are example of :-

(1) Adaptive radiation
(2) Adaptive convergence
(3) camouflage
(4) Mimicry

84) In the given table identify the pair which shows convergent evolution :-

	Column-I	Column-II
(a)	 Bobcat	 Tasmanian wolf
(b)	 Lemur	 Spotted cuscus
(c)	 Anteater	 Tasmanian tiger cat
(d)	 Flying squirrel	 Flying phalanger

(1) a, b
(2) b, c
(3) b, d
(4) c, d

85) The big-bang theory attempts to explain to us :-

(1) Origin of life
(2) Origin of plants
(3) Origin of universe
(4) Origin of life from unknown planet

86) Chemical evolution is :-

(1) Formation of inorganic molecules from organic molecules.
(2) Formation of organic molecules from inorganic constituents.
(3) Formation of recent bio diversity from first life.
(4) Origin of first life from non living organic molecules.

87) Choose the incorrect statement for *Homo sapiens neanderthalensis*

(1) Fossil of human
(2) Large brain with 800-900 cc
(3) Lived between 1,00,000-40,000 years back
(4) Buried their dead relatives

88) **Assertion :** The process of evolution of different species in a given geographical area starting from a point and literally radiating to other areas of geography (habitats) is called adaptive radiation.
Reason : When more than one adaptive radiation appeared to have occurred in an isolated geographical area (representing different habitats), one can call this convergent evolution.

(1) Both Assertion and Reason are True and the Reason is a correct explanation of the Assertion.
(2) Both Assertion and Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion and Reason are False.

89)

During evolution which organism deposits and convert in to present days coal :-

(1) Angiosperms
(2) Dicotyledons
(3) Bryophyte
(4) Pteridophytes

90) Which human ancestor painted beautiful paintings on cave wall and developed pre historic cave art 18000 years ago ?

(1) *Homo habilis*
(2) Cromagnon man
(3) Java man
(4) *Australopithecus*

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	1	3	2	1	1	2	1	3	3	3	1	4	3	2	3	1	1	1	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	3	1	4	1	3	4	1	3	1	3	2	2	1	4	3	3	2	2	2
Q.	41	42	43	44	45															
A.	3	2	4	2	1															

CHEMISTRY

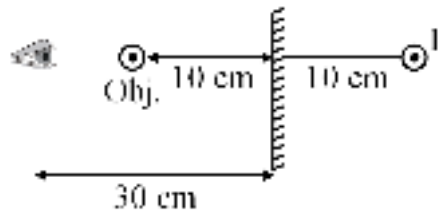
Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	1	1	1	3	2	2	1	2	1	3	3	3	2	1	1	1	3	4	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	4	2	3	2	2	2	3	1	3	4	3	4	4	1	4	2	1	1	1
Q.	86	87	88	89	90															
A.	3	1	2	4	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	3	1	4	2	1	4	2	4	3	3	1	1	2	3	1	4	1	2	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	4	1	2	1	4	1	2	4	3	1	3	3	3	2	2	2	4	3	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	4	4	3	1	2	3	1	2	3	1	3	3	4	3	4	2	2	3	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	1	3	4	4	1	4	3	4	1	1	2	4	2	3	1	4	1	2	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	1	1	3	3	2	2	2	4	2										

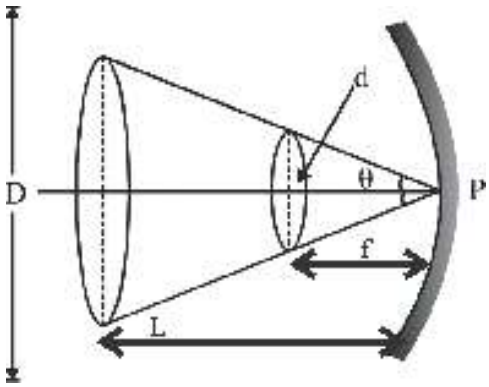
SOLUTIONS

PHYSICS



1)

Focused our eye at 40 cm



2) The image will be formed at focus because object is at large distance.

The angle subtended by diameter of moon at pole

$\tan\theta = \frac{D}{L}$, since θ is small, $\tan\theta = \theta = \frac{D}{L}$ radian

By similar triangles

$$\theta = \frac{d}{f} = \frac{D}{L}$$

$d = f\theta = f\frac{D}{L} = 7.6 \times \frac{3450}{3.8 \times 10^5} = 0.069$ m = 6.9 cm

d; diameter of image of moon.

3) $\mu_1 \sin i = \mu_2 \sin r$

$1 \times \sin i = \mu \sin(i/2)$ {i = 2r}

$$\cos(i/2) = \frac{\mu}{2}$$

$$i = 2\cos^{-1}\left(\frac{\mu}{2}\right)$$

4)

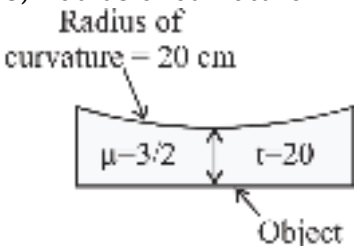
If object is moved away from mirror, its image moves towards focus, so size decreases.

5) $$\mu = \frac{\text{Real depth}}{\text{apparent depth}} = \frac{120}{80} = 1.5$$

6) TIR is possible when light goes from denser to rarer medium, and incidence angle is greater than critical angle.
$i > \theta_C$

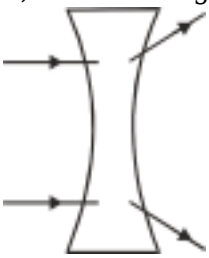
7) 1sini = μsinr
$\sin r = \frac{1}{\mu}\sin i$
$\mu = a + \frac{b}{\lambda^2}$ and λ_v is minimum and λ_R is maximum.

8) Radius of curvature = 20 cm



Considering refraction at the curved surface,
$u = -20$; $\mu_2 = 1$
$u_1 = 3/2$; $R = +20$
Applying $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$
$\frac{1}{v} - \frac{3/2}{-20} = \frac{1-3/2}{20} = v = -10$
i.e. 10 cm below the curved surface or 10 cm above the actual position of flower.

9) It will diverge a parallel beam if $\mu_{inside} > \mu_{outside}$



For (1) : $\mu_{inside} = \mu_2 > \mu_{outside} = \mu_1$
For (2) : $\mu_{inside} = \mu_2 > \mu_{outside} = \mu_\omega$
For (3) : $\mu_{inside} = \mu_\omega > \mu_{outside} = \mu_1$
For (4) : $\mu_{inside} = 1 < \mu_{outside} = \mu_1$ or μ_2 or μ_ω

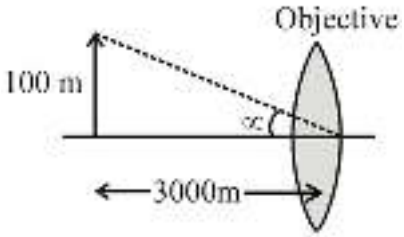
10) Original aperture diameter = d
Original area ∝ d^2Blocked central diameter = d/3
Blocked area ∝ $(d/3)^2 = d^2/9$Fraction of area blocked = $(d^2/9) / d^2 = 1/9$
Therefore fraction of light passing = 1 − 1/9 = 8/9New intensity = (8/9)I

11) $m = m_0 \times m_e$m
$= m_0 \times \left(1 + \frac{D}{f_e}\right)$

$\Rightarrow 100 = 10 \times \left(1 + \frac{25}{f_e}\right) \Rightarrow f_e = \frac{25}{9}\text{cm}$

12)

This is the case of V.O & R.I
u = +10 cm
V = 15 cm
$\frac{1}{V} - \frac{1}{u} = \frac{1}{f}$
$\frac{1}{15} - \frac{1}{10} = \frac{1}{f}$
f = –30cm



13)

$\tan \propto \approx \alpha = \frac{100}{3000}$

$\propto = \frac{1}{30}\text{rad}$

M.P. $= \frac{f_0}{f_e} = \frac{\beta}{\propto}$

$\frac{120}{5} = \frac{\beta}{1} \times 30$

$\beta = \frac{4}{5} = 0.8$ radian

14)

Apparent depth of the object $= \frac{33.25}{1.33}\text{cm}$

Distance of object from the mirror $= 15 + \frac{33.25}{1.33}$

$= 40\text{cm}$

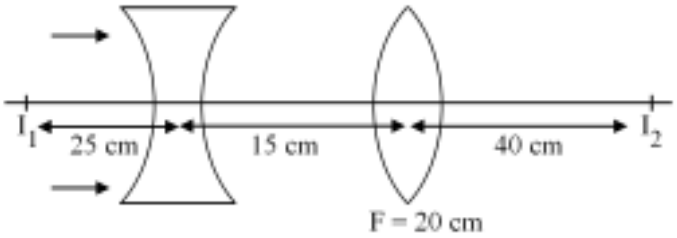
Similarly, distance of image from the mirror

$= 15 + \frac{25}{1.33} = 33.8\,\text{cm}$

For concave mirror, u = –40cm or v = –33.8cm
Applying mirror formula,

$f = \frac{uv}{u+v} = \frac{(40)(-33.8)}{(-40-33.8)} = 18.3\,\text{cm}$

15) Shift $= \left(t_1 - \frac{t_1}{\mu_1}\right) + \left(t_2 - \frac{t_2}{\mu_2}\right) + \left(t_3 - \frac{t_3}{\mu_3}\right) + \left(t_4 - \frac{t_4}{\mu_4}\right)$



16)
Real image

17)

(A) $Y = \frac{\lambda D}{d} = 1^{st}$ maximum $= 4I^0$

(B) $Y = \frac{\lambda D}{2d} = 1^{st}$ minimum $= 0$

(C) $Y = \frac{\lambda D}{3d}$

$\Delta x = \lambda/3$

$\Delta\phi = 2\pi/3$

$I_{Res} = 4I_0\cos^2(\pi/3) = I_0$

(D) $Y = \frac{\lambda D}{4d}$

$\Delta x = \lambda/4$

$\Delta\phi = 2\pi/4 = \pi/2$

$I_{Res} = 4I_0\cos^2(\pi/4) = 2I_0$

18) $(\mu - 1)\,t \cdot \frac{\beta}{\lambda} = \beta$

$\mu - 1 = \frac{\lambda}{t} \Rightarrow \mu = 1 + \frac{\lambda}{t}$

$= 1 + \frac{\lambda}{t} = 1 + \frac{60}{120} = 1.5$

19)

Position of I^{st} minima

$y = \frac{\lambda D}{a} = \frac{(5 \times 10^{-8})(0.6)}{0.02 \times 10^{-2}}$

y = 0.15 cm

20) After 1 polariser the intensity is

$I^1 = \frac{2I_0}{2} = I_0$

After 2 Polarisor

$I^{11} = I_0\cos^2\theta$

After 3 polariser

$I^{111} = I_0\cos^2\theta\cos^2(90-\theta) = I_0[\cos\theta\sin\theta]^2$

$= \frac{I_0}{4}[2\sin\theta\cos\theta]^2 = \frac{I_0}{4}\sin^2 2\theta$

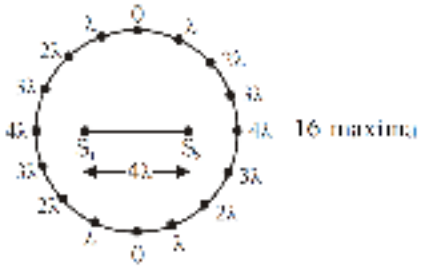
21)

In Brewster law the angle between reflected and refracted rays are mutually perpendicular.

22)

$I_0 = I + I + 2\sqrt{I I}\cos\phi$ | $\because I_0 = I$
$I = 2I(1 + \cos\phi)$
$I = 2I \times 2\cos^2 \phi/2$
$\phi = 120° = 2\pi/3$

23) $\lambda = 0.5$ m
d = 2m
So distance between S_1 & S_2 $(0.5\lambda) \times 8$
$= 4\lambda$



24) The density of nuclear matter is approximately constant and independent of the size or mass number of the nucleus. All nuclei whether they are the original parent nucleus or the daughter nuclei formed after disintegration have same density. So, the density of the smaller nucleus will be the same as the original nuclear density ρ.

25)

Statement-I : "Saturation of nuclear force is the reason for the constancy of the binding energy per nucleon."
Correct, because each nucleon interacts only with nearest neighbors. This leads to Binding Energy (BE) ∝ Mass Number (A) for a nuclei, which means BE/A roughly constant.
Statement-II : "The nuclear binding force has to dominate over the Coulomb repulsive force between protons inside the nucleus, to make the nucleus stable."
Correct, because for nucleus stability, attractive nuclear force must overcome repulsive Coulomb force between protons. Otherwise, the protons will not remain inside the nucleus.
Thus, both the statements are correct.

26) ${}^{A}_{Z}X \xrightarrow{\beta^+} {}_{Z-1}B \xrightarrow{\alpha} {}_{Z-3}C \xrightarrow{\beta^-} {}_{Z-2}D$
β^+ decreases atomic number by 1
α decreases atomic number by 2

β^- increases atomic number by 1

27) $r_n = r_0 \frac{n^2}{Z} \times \frac{m_e}{m}$

$= 0.529 \times 10^{-10}\ m \times \frac{(1)^2}{1} \times \frac{m_e}{50 m_e}$

$r \approx 1.058 \times 10^{-12}\ m$

$E_n = E_1 \times \frac{Z^2}{n^2} \times \frac{m}{m_e}$

$= -13.6\,eV \times \frac{(1)^2}{(1)^2} \times \frac{50 m_e}{m_e}$

$E_1 = -13.6\ eV \times 50 = -0.68\ keV$

28) Transition to n = 1 and n = 2 are referred to as Lyman and Balmer series respectively.
For Lyman series, wavelength will be longest when the electron has a transition from n = 2 to n = 1 level.
For Balmer series wavelength will be longest when electron has a transition from n = 3 to n = 2 level.

as $E = \frac{1}{\lambda}$

$\frac{1}{\lambda_L} = R\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = \frac{3R}{4}$

$\frac{1}{\lambda_B} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = \frac{5R}{36}$

$\frac{\frac{1}{\lambda_L}}{\frac{1}{\lambda_B}} = \frac{\frac{3}{4}}{\frac{5}{36}} = \frac{27}{5}$

Hence, $\frac{\lambda_L}{\lambda_B} = \frac{5}{27}$

29) (X) → (Y) + (Z) + (P)
M M/3 M/3 M/3

$\Delta Mc^2 = \frac{1}{2}\frac{M}{3}V^2 + \frac{1}{2}\frac{M}{3}V^2 + \frac{1}{2}\frac{M}{3}V^2$

$V = c\sqrt{\frac{2\Delta M}{M}}$

30) $P = \left(\frac{N}{t}\right)$ (energy in each fission)

31)

The frequencies are :

$\nu_1 = \frac{\omega_1}{2\pi} = \frac{5 \times 10^{15}}{2\pi} \approx 7.96 \times 10^{14}$ Hz

$\nu_2 = \frac{\omega_2}{2\pi} = \frac{8 \times 10^{15}}{2\pi} \approx 12.73 \times 10^{14}$ Hz

Energy of photons for each frequency

$E = h\nu$

$E_1 = \frac{6.6 \times 10^{-34} \times 7.96 \times 10^{14}}{1.6 \times 10^{-19}} \approx 3.29$ eV

$E_2 = \frac{6.6 \times 10^{-34} \times 12.73 \times 10^{14}}{1.6 \times 10^{-19}} \approx 5.265$ eV

Maximum kinetic energy will be determined by the higher energy photons (E_2)

$K_{max} = E - \phi$

$= 5.265$ eV $- 2$eV

$K_{max} \approx 3.27$ eV

32) 4×10^3 J $= (10^{20})\ E_{photon}$

$E_{photon} = \frac{4 \times 10^{-17}\ J}{1.6 \times 10^{-19} J}.eV = \frac{1}{0.4} \times 10^2$ eV

$\frac{hc}{\lambda} = 250$ eV $\Rightarrow \quad \lambda = \frac{12400}{250}$eV

$\lambda \approx 50$ Å

33)

The work function ϕ can be calculated using Einstein's photoelectric equation.

For initial wavelength : $5 = \frac{hc}{\lambda} - \phi$ .....(1)

For decreased wavelength : $17 = \frac{hc}{0.4\lambda} - \phi$ .....(2)

Eliminating $\frac{hc}{\lambda}$ from (1) and (2), we get :

$\phi = 3$ eV.

34)

De-Broglie wavelength of electrons

$\lambda = \frac{h}{mv} \Rightarrow m = \frac{h}{v\lambda}$

Kinetic energy of electron

$K_e = \frac{1}{2}mv^2 = \frac{1}{2} \times \frac{h}{v\lambda} \times v^2 = \frac{hv}{2\lambda}$ ... (i)

Energy of photon, $E_P = \frac{hc}{\lambda}$ ... (ii)

$\frac{K_e}{E_P} = \frac{hv/2\lambda}{hc/\lambda} = \frac{v}{2c}$

$= \frac{1.5 \times 10^8}{2 \times 3 \times 10^8} = \frac{1}{4}$

35)

$$\lambda = \frac{h}{mv} = \frac{h}{qrB}$$

$$\Rightarrow \frac{\lambda_\alpha}{\lambda_{pr}} = \frac{q_{pr} \times r_{pr}}{q_\alpha \times r_\alpha} = \frac{1}{2}$$

36) The diode allows only the positive half cycle of AC to pass.
The capacitor charges to the peak positive voltage (+5 V) and holds it (since discharge through 1k is slow compared to input period).
Thus, output is nearly a constant DC voltage close to +5 V.

37) Impurity is trivalent (In)
 using $n_e n_h = n_i^2$
$(1.2 \times 10^9)\ n_h = (6 \times 10^{15})^2$

$$n_h = \frac{36 \times 10^{30}}{1.2 \times 10^9} = 3 \times 10^{22} m^{-3}$$

38) $Y = \overline{\bar{A} \cdot \bar{B}} = \overline{\bar{A}} + \overline{\bar{B}} = A + B = \text{OR gate}$

39) $$R_F = \frac{\Delta V}{\Delta I} = \frac{0.1}{400 \times 10^{-3}} = 0.25\Omega$$

40) $$I_L = \frac{V_Z}{R_L} = \frac{10}{5k\Omega} = 2mA$$

$$I = \frac{V - V_Z}{R_S} = \frac{24 - 10}{1k\Omega} = 14mA$$

$I_Z = I - I_L = 14 - 2 = 12$ mA
$P_Z = V_Z\ I_Z = 10 \times (12 \times 10^{-3}) = 120$ mW

41)

In F.B. diode allow current in mA.

42) $$\frac{1}{f + 20} - \frac{1}{-(f + 20)} = \frac{1}{f} \Rightarrow f = 20cm$$

43) $d \sin\theta = \left(n - \frac{1}{2}\right)\lambda$

$$\sin\theta = \left(n - \frac{1}{2}\right)\frac{\lambda}{d} \leq 1$$

$$\left(n - \frac{1}{2}\right) \leq \frac{d}{\lambda}$$

$n - \frac{1}{2} \le \frac{4\lambda}{\lambda}$

$n \le 4{\cdot}5$

$\Rightarrow$ n = 1, 2, 3, 4

$\Rightarrow$ Total min = 2(4) = 8

44) For A to C

$A_C = A_1 - 3 \times 4 = A_1 - 12$ ...(1)

For B to C

and $A_C = A_2 - 2 \times 4 = A_2 - 8$ ...(2)

by eq (1) & (2) $A_1 - 12 = A_2 - 8$

$\Rightarrow A_1 - A_2 = 4$ ...(3)

Again for A to C

$Z_C = Z_1 - 3 \times 2 + 1 = Z_1 - 5$ ...(4)

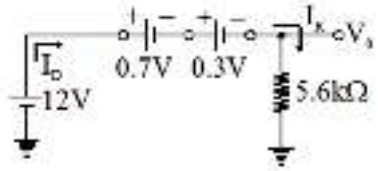
For B to C, $Z_C = Z_2 - 2 \times 2 + 5 = Z_2 + 1$ ...(5)

by (4) & (5)

$Z_1 - 5 = Z_2 + 1$

$Z_1 - Z_2 = 6$ ...(6)

45) Direction of current I_D is in the same direction as arrow in the diode symbol. Both the diodes are in "on" state; hence voltage drops across their terminal V_{T_1} and V_{T_2} are 0.7 V and 0.3 V respectively. The network function because E = 12 V > (0.7 V + 0.3 V).



$V_0 = E - V_{T_1} - V_{T_2} = 12 - 0.7 - 0.3 = 11$ V.

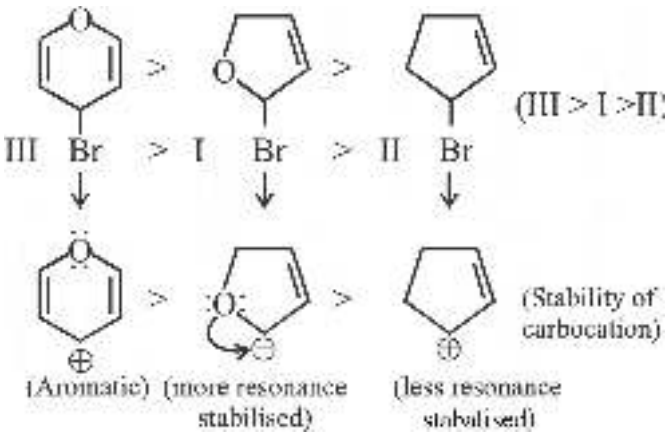
CHEMISTRY

46)

Leaving group tendency $\propto$ Stability of anion

47)

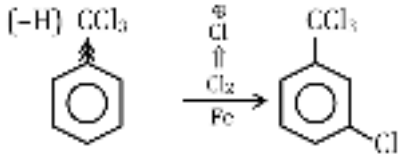
Reactivity towards $SN^1 \propto$ Stablity of intermediate carbocation.



Final answer : 1

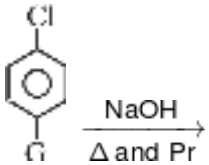
48)

S_{N^2} reaction, inversion take place

49)  **Final Answer :**

Option (1)

50)

 Here 'G' must be -M group to increase rate of NSR in Aryl Halides.

 G ⇒ $-NO_2$ [-M group]

Final answer : Option (3)

51)

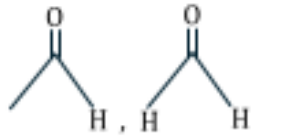
A. Reaction (a) is dehydrohalogenation. It is elimination reaction. Removal of HBr takes place from substrate and alkene is formed.
B. Reaction (b) is nucleophilic substitution reaction. Bromine is substituted by hydroxyl group bromo alkane is converted to alcohol during nucleophilic substitution reaction.
C. Reaction (c) is addition reaction Br_2 is added across pi bond of alkene.

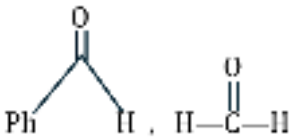
Final Answer:

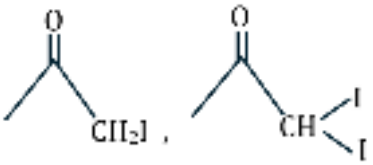
The correct option is (2)

52)

(1) 
Both give iodoform test
can't distinguish

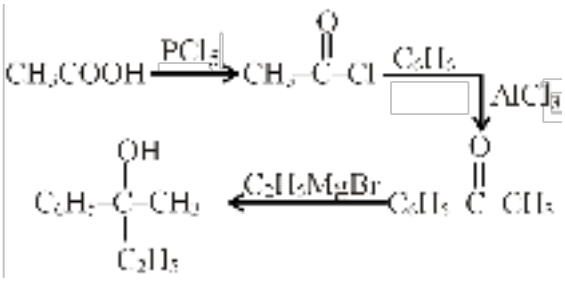
(2) 
give test Not give test
distinguish by Iodoform test

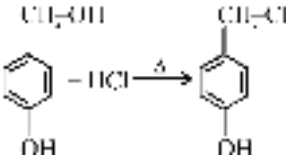
(3) 
Both does not give test
can't distinguish

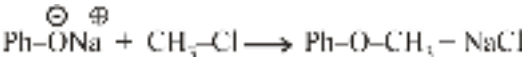
(4) 
Both give iodoform test can't
distinguish.

Final Answer : option (2)

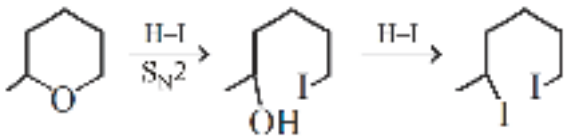
53) NCERT XII Part-II, Edition (2023-24) Pg.169
Boiling point and molecular mass

54) 

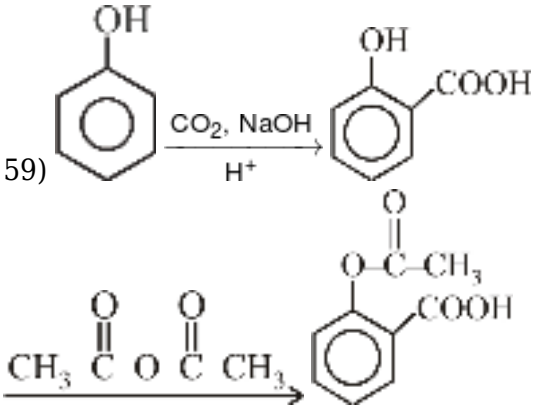
55) 

56) 

57) NCERT-XII Pg#367

58) 

Final Answer: The final product is a di-iodide

59) 

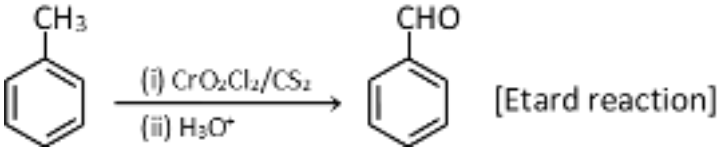
60)

3° Alcohol $\xrightarrow[\text{(Lucas reagent)}]{\text{anhy } ZnCl_2 + HCl}$ Immediate Turbidity

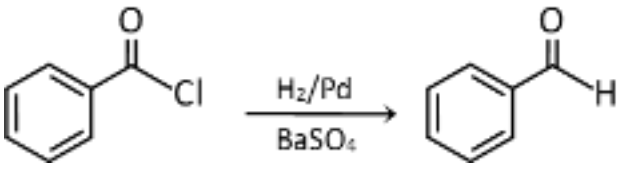
1° Alcohol $\xrightarrow[\text{Room Temperature}]{HCl}$ No reaction

61) **Solution :** Ethanol's molecules form hydrogen bonds with each other due to the presence of the –OH group.
Dimethyl ether cannot form intermolecular H-bonds (only weak dipole-dipole interactions).
Therefore, ethanol has a higher boiling point than dimethyl ether.

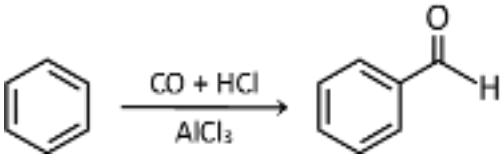
62) **Solution :**

(1) 

[Etard reaction]

(2) 

[Rosenmund reduction]

(3) 

[Gatterman Koch reaction]

(4) $CH_3CN \xrightarrow[(ii)\ H_3O^+]{(i)\ SnCl_2 + HCl} CH_3CHO$ [Stephen's reaction]

Hence,
Option (1) is correct

63)

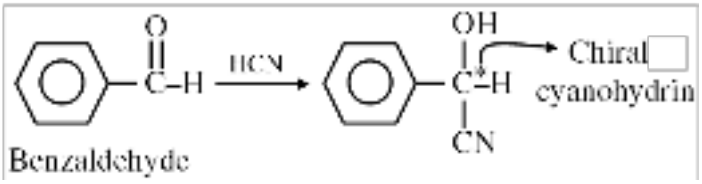
Solution : Reactivity for Nucleophilic

Addition $\propto \frac{1}{\text{Steric Hindrance}} \propto \frac{1}{\text{Electron} - \text{Donating Effect } (+I/+M)}$

Aldehyde is more reactive than ketones by both factors.

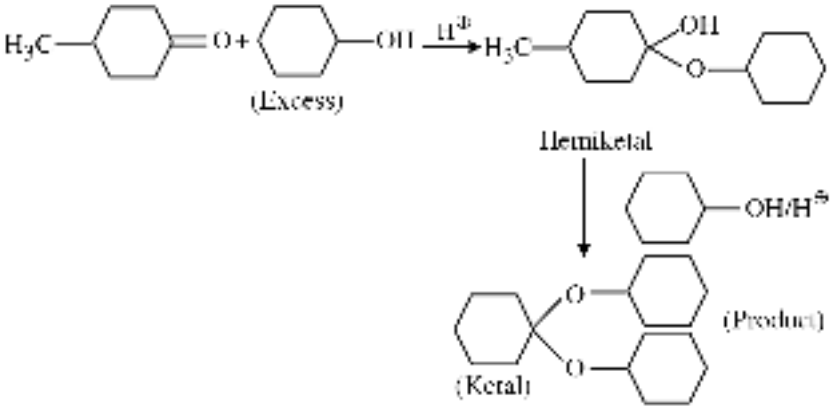
64)

Solution :-



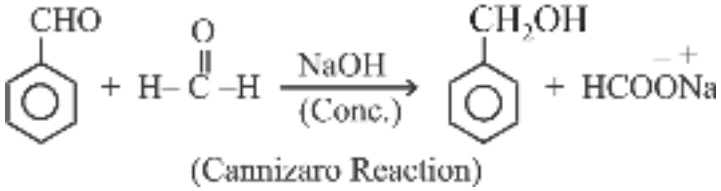
65)

Solution :-



Final Answer :- 3

66)

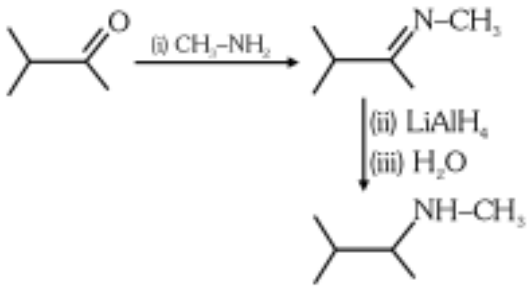


67) 1. $CH_3CHO + CH_3CHO \rightarrow CH_3CH=CHCHO$ (But-2-enal)
2. $CH_3CH_2CHO + CH_3CH_2CHO \rightarrow CH_3CH_2CH=C(CH_3)CHO$ (2-Methyl-pent-2-enal)
3. $CH_3CH_2CHO + CH_3CHO$ (enolate) $\rightarrow CH_3CH_2CH=CHCHO$ (Pent-2-enal)

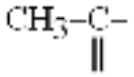
4. 3. CH_3CH_2CHO(enolate) + CH_3CHO → $CH_3CH=C(CH_3)CHO$ (2-Methyl-but-2-enal)

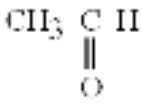
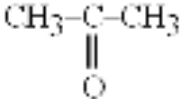
68)

Solution :-

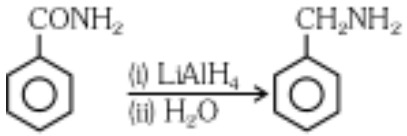


Final answer :- (2)

69) **Solution :** Methyl ketone () containing compounds can give iodoform test.

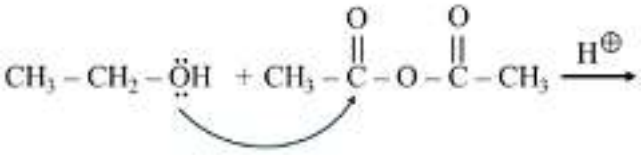
Ex.  and 

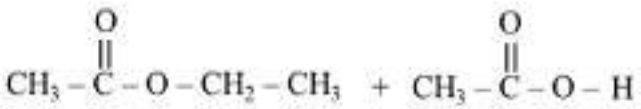
Final answer : 3

70) 

71)

Solution:





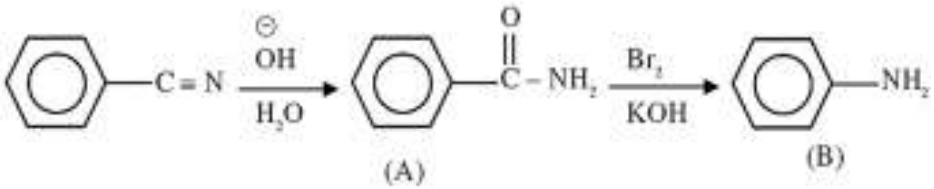
Nucleophilic substitution

72)

NCERT-XII, Aldehyde, Ketone & Carboxylic acid

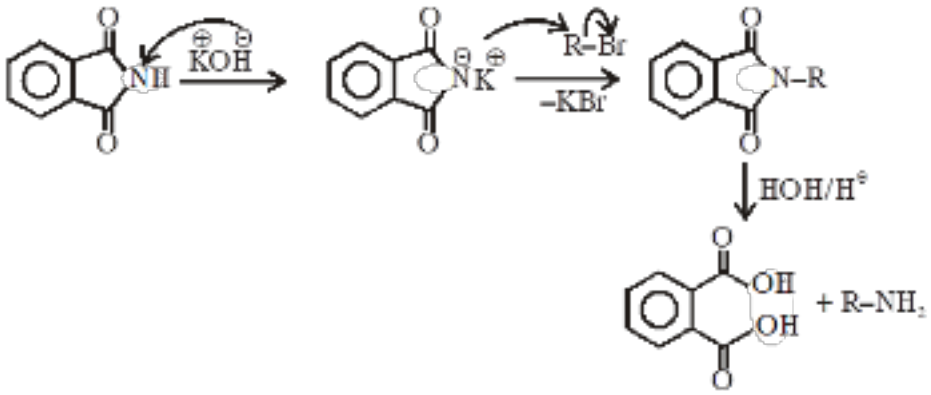
73)

Solution :-

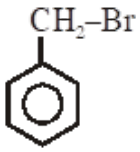


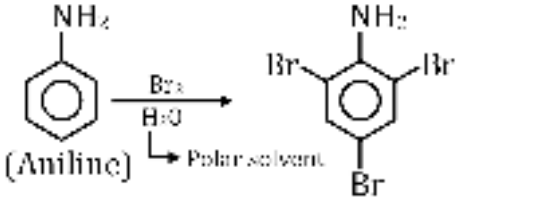
The product 'B' is Aniline.

Final Answer :- 3



74) Here R–Br =

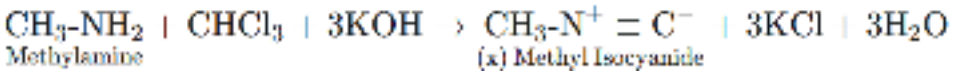




75)

Note:- Since, $-NH_2$ is a highly Activating group it is O & P directing and we get trisubstituted aniline.

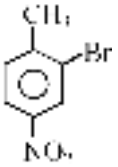
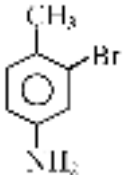
76)

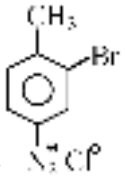
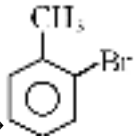


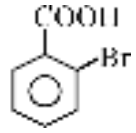
77)

Compound	Structure	Amine classification	Product with HNO_2
(1)	CH_3NH_2	1°	N_2 gas (Bubble formation)

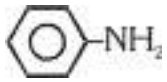
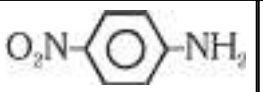
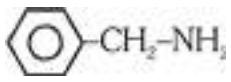
(2)	$CH_3CH_2NH_2$	1°	N_2 gas (Bubble formation)
(3)	$CH_3CH_2NHCH_2CH_3$	2°	Oily Nitrosoamine
(4)	$(CH_3CH_2)_3N$	3°	No visible reaction

78) (A)→  (B)→ 

(C)→  (D)→ 

(E)→ 

79)

Compounds	Types of amine	Diazotization Y/N
1 	Aromatic 1° amine	Yes
2 	Aromatic 1° amine	Yes
3 	Aromatic 1° amine	Yes
4 	Aromatic 1° amine (benzylamine)	No

Final Answer : Option (4)

80) **Solution:**

Vitamin A deficiency causes night blindness and Xerophthalmia (dryness of the conjunctiva and cornea).

Vitamin C deficiency causes Scurvy, which is characterized by bleeding gums.

Vitamin B1 deficiency causes the disease Beri-Beri, which affects the nervous and cardiovascular systems.

Vitamin D deficiency causes Rickets in children and osteomalacia in adults, leading to soft and

weak bones.
A-iii, B-i, C-iv, D-ii.
Final Answer: (1)

81) **Solution :-**
(1) protein are most abundant biomolecule in living system
(2) cellulose is the most abundant arganic substance in plant kingdom → correct
(3) glucose is most abundent organic compound]
on earth → correct.
Final answer :- 4

82) Here, α–D-(+) glucose and β–D-(+) glucose differ only at the anomeric carbon (C1) → they are anomers.
Final Answer: Option (2)

83) Key Rule: Cyclic acetals masks the aldehyde; anomers give α/β forms.
• Statement-I: Correct — Pentaacetable lacks free CHO, so no oxime with H_2N-OH.
• Statement-II: Correct — Glucose has α and β forms due to the anomeric OH at CI.
Simple reaction: Glucose pentaacetate +H_2N-OH-> No reaction (no free CHO).
Final answer: Option (1)

84) Thyroxine hormone contain iodine

85) α-D glucopyranoside + β-D-fructofuramoside
Sucrose ↲

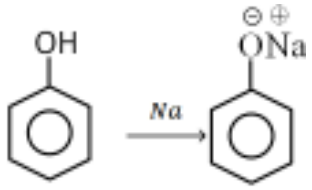
86)

t-Butyl chloride ($(CH_3)_3C$–CI): SN1 reactivity (tertiary halide).
Forms white precipitate (AgCI) readily.
Chlorobenzene (C_6H_5CI): Aryl chloride (C-Cl bond is resonance-stabilized).
Does not react with $AgNO_3$ (no precipitate) because the C–Cl bond is too strong to ionize.
Benzyl chloride ($C_6H_5CH_2Cl$): Highly reactive (benzylic C-Cl bond).
Forms white precipitate (AgCl) instantly.
Final Answer :- Option 1

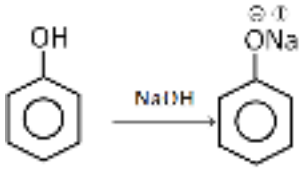
87) NCERT XII II part Pg.# 355

88) **Answer Explanation :**
In sodium fusion, nitrogen in the organic compound reacts with sodium and carbon to form sodium cyanide (NaCN).$NaNO_2$, $NaNH_2$, NaNC are not formed in this process.
Final Answer : Option (2)

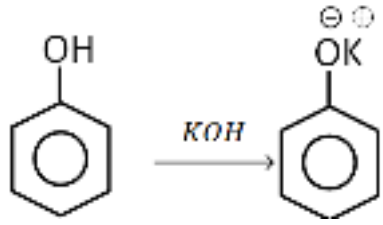
89)

1 . 

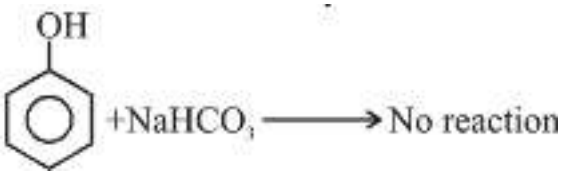
(strong acid)

2. 

+ H_2O (weak acid)
(strong acid)

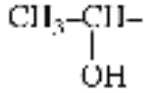
3. 

+ H_2O (weak acid)
(strong acid)

4. 

because phenol is weaker acid then H_2CO_3
Final answer: option (4)

90)

• CAN test for alcohol
• 3° ROH immidiate turbidity
2° ROH turbidity after 5 min
• Iodoform test : 

BIOLOGY

91)

NCERT XII, Pg. # 202
Question Explanation: The question is asking: “Why does the fig wasp enter the fig inflorescence?”
Concept: Mutualism
Solution: Explanation:Fig wasps enter the fig inflorescence to lay their eggs (oviposition) inside some of the flowers.
While doing so, they accidentally transfer pollen, thereby pollinating the fig.

This is a classic example of obligate mutualism: the fig depends on the wasp for pollination, and the wasp depends on the fig for a site to lay eggs.
Final answer: The correct answer is option 2. Oviposition/egg-laying

92) NCERT, Pg. # 201

93)

NCERT XII, Pg.# 200
Question Explanation: The question is asking: Which organism needs two intermediate hosts (snail and fish) to complete its life cycle?

Concept: Parasitism

Solution:

Reason:

The human liver fluke requires two intermediate hosts:

-Snail
-Fish

Then it infects humans.

Final answer: The Correct answer: 1. Human liver fluke

94) NCERT Pg. # 200

95)

NCERT XII Pg.233-234

96)

NCERT XII, Pg.# 197

97)

NCERT XII, Pg.# 201

98)

Module

99) NCERT Page No. 198

100) NCERT, Pg. # (E)-197, (H)-215

101) NCERT Pg.# English- 198,199.
Hindi- 217, 218.

102)

NCERT-XII, Pg. # 198

103) NCERT Pg. No.#197, 199, 200, 201.

104) NCERT Page No. 228

105) NCERT, Pg. # 227

106) NCERT Pg. # 209

107) NCERT, Pg. # 209

108) NCERT Pg. # 246

109) NCERT Pg. # 213

110)

Module

111) NCERT Pg. # 207

112) NCERT Pg. # 209

113) NCERT Pg. 207

114) NCERT, Pg. # 211,206

115)

NCERT-XII, Pg. # 206

116) NCERT, Pg. # 209,206

117) NCERT, Pg. # 209

118) NCERT Pg. # 213

119) NCERT Pg. # 205 & 209

120)

NCERT Pg. # 207 - 210

121) NCERT XII, Pg. # 216-217

122)

NCERT XII Pg : 260

123) NCERT-XII, Pg. # 219
Question Explanation: Number of bird species in India.
Concept: Indian biodiversity.
Solution: India has more than 1200 species of birds.
Final Answer: Option (3)

124) NCERT-XII, Pg. # 221
Question Explanation: Scientist who showed diversity leads to higher productivity.
Concept: Diversity-productivity hypothesis.
Solution: David Tilman showed in his experiments that increased diversity contributed to higher productivity.
Final Answer: Option (3)

125) NCERT Page. No : 223

126) NCERT Page. No : 223

127) NCERT Page. No : 223

128)

NCERT-XII, Pg. # 266,267

129) NCERT Page No . 225

130)

Module

131)

NCERT-XII, Pg.# 218

132) NCERT, Pg # 217, 218

133) XII-NCERT Page No. # 263, 264

134) NCERT, Pg. # 223-224

135) NCERT XII # 263, 264

136)

NCERT (XIIth) Pg. # 136

137) NCERT, Pg. # 146, 147

138)

NCERT, Pg. # 141

139) NCERT-(XII) Pg # 150

140) NCERT Pg. # 147

141)

NCERT, Pg. # 130

142)

NCERT, Pg. # 142, 143

143) NCERT Page # (E) 149, (H) 160

144)

NCERT, Pg. # 137

145)

Module

146)

NCERT-XI, Pg. # 160

147)

NCERT, Pg. # 130

148) NCERT Pg. # 133

149) NCERT-XII, Pg. # 164, 153

150)

NCERT, Pg. # 132

151)

NCERT, Pg. # 145

152) NCERT Pg. # 135 (E), 148 (H)

153) NCERT–XII, Pg. # 158,159, Para-8.5

154) NCERT Pg. # 151

155) NCERT XIIth Pg.#159

156)

NCERT, Pg. # 143

157) NCERT-XII Pg. # 136

158) NCERT Page No. 134

159)

NCERT, Pg. # 145

160) Module-11 Page No. #06

161)

Module-11 Page No. # 09

162) NCERT-XII, Pg. # 144

163)

NCERT-XII, Pg.# 137

164)

NCERT, Pg. # 116

165)

NCERT, Pg. # 111, 112

166) NCERT XI Pg # 131

167)

NCERT-XII, Pg. # 116

168)

NCERT, Pg. # 122

169)

Module-11 Page No. #107

170)

NCERT, Pg. # 124

171)

Module

172) NCERT Pg # 138

173)

NCERT, Pg # 117

174)

NCERT Pg # 118

175) NCERT XII, Page # 127

176) NCERT XII, Page # 127

177) NCERT Page-141

178) NCERT Pg. 117, 118

179)

NCERT, Pg. # 122

180) NCERT XIIth Pg. # 125